

Indian Institute of Engineering Science and Technology, Shibpur



Department of Physics

Practical Report

PH6272: Special Laboratory for Nuclear and Particle Physics

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Experiment 1

Detection of gamma radiation of known sources using Sodium Iodide detector and single channel analyzer (SCA)

1.1 Objective

1. To find the energy resolution of the detector as a function of gamma energy.

1.2 Theory

Gamma-ray detection using scintillation detectors like sodium iodide (NaI) relies on the scintillation phenomenon, which involves the conversion of gamma-ray energy into visible light. Here are the key principles behind gamma-ray detection using NaI scintillation detectors:

Scintillation Material: Sodium iodide (NaI) is a commonly used scintillation material for gamma ray detection. NaI crystals are doped with small amounts of thallium (Tl) to enhance their scintillation efficiency. When gamma rays interact with the NaI crystal, they deposit energy in the material, causing the atoms to become excited.

Excitation and De-Excitation: The energy deposited by gamma rays excites the atoms in the NaI crystal to higher energy levels. As these excited atoms return to their ground state, they emit photons of visible light in a process known as scintillation. The intensity of the scintillation light is proportional to the energy deposited by the gamma rays in the crystal.

Photon Detection: The scintillation light produced in the NaI crystal needs to be detected and converted into an electrical signal for further analysis. This is achieved using photomultiplier tubes (PMTs) or other light-sensitive detectors coupled to the crystal. PMTs are particularly

suited for this purpose due to their high sensitivity and ability to amplify weak light signals.

Photomultiplier Tubes (PMTs): Photomultiplier tubes are vacuum tubes that contain a photosensitive cathode, a series of dynodes, and an anode. When a photon strikes the cathode, it releases photoelectrons, which are accelerated towards the dynodes by an electric field. Each dynode is at a higher potential than the previous one, causing electron multiplication through secondary emission. This cascade amplification process results in a significant increase in the number of electrons, leading to an electrical pulse at the anode. The photoelectrons are ejected from the photocathode into the vacuum with energy of $\approx 1\text{eV}$ and are accelerated by a voltage of a few hundred volts toward the dynode through a focusing electrode. So, the accelerated electron has energy of a few hundred electron volts upon arrival at the 1st dynode. This deposition of kinetic energy can result in the re-emission of secondary electrons. It typically takes 2-3 eV to excite an electron in the dynode, so 100 V can theoretically create ≈ 30 electrons. Also, electrons must be near the surface to be emitted, otherwise they lose too much energy through collisions and are unable to overcome the potential barrier at the surface. Now, the secondary electrons from the first dynode are accelerated by the electric field toward a second dynode, where they create secondary electrons, and so on. In each step there is a certain amount of multiplication for each of the incident electrons called the overall multiplication factor defined by

$$d = \frac{\text{Number of secondary electron transmitted}}{\text{primary incident electron}} \quad (1.1)$$

The overall gain for the PM tube = $a \times d^N$ where **N** is the number of multiplication stages, **a** is the fraction of electrons collected by the multiplier structure, and **d** is the multiplication factor for **a** single electrode. For typical tubes **d** is around 5 and **a** approaches unity. 10 stages of multiplication result in an overall tube gain of 5^{10} . Large value of **d** is required for reducing statistical fluctuations. Of course, we have to adjust the gain according to the strength of the radioactive source.

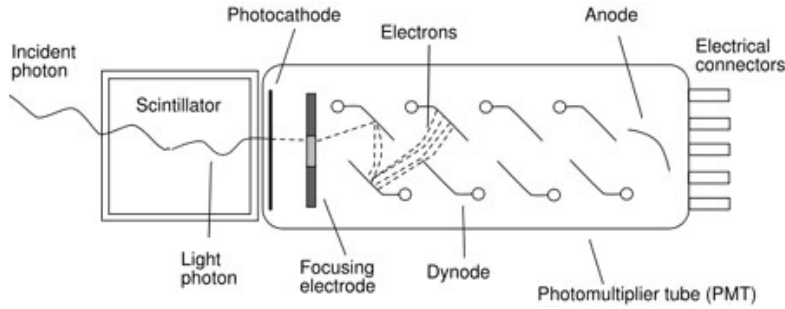


Figure 1.1: Schematic diagram of a scintillation detector comprising a scintillation material coupled to a photomultiplier tube.

Details of the sources

Co^{60} : The most probable decay chain of Co^{60} is first decaying to an excited state of Ni^{60} by beta emission and then to the ground state of Ni^{60} by emitting two subsequent gamma ray photons of energies 1173.2 keV and 1332.5 keV. The half-life of Co^{60} is 5.272 years. Now our source is shielded source that absorbs all the beta rays, so we can detect only the gamma rays.

Cs^{137} The most probable decay chain of Cs^{137} is first decaying to an excited state of Ba^{137} by beta emission and then decaying to the ground state of Ba^{137} by a gamma emission of energy 662 keV. The half-life of Cs^{137} is about 30 years.

Experimental Procedure

We are taking the data in **Threshold mode** and **Window mode**. In both the cases the signal just after emerging from the PMT is amplified by a preamplifier and then the passing through a cable the signal is amplified to a sufficient value in the nuclear spectrometer and then it is displayed. The requirement of the preamplifier is very important as it amplifies the signal at its initial stage so that throughout its long journey through the cable the Signal to Noise Ratio(SNR) is maintained at a sufficient value.

Threshold mode

In the threshold mode we have basically given a lower threshold voltage beyond which every signal generating a voltage upto 10.00 volt (upper limit of the counter) is recorded and we have shifted the threshold voltage in steps of 0.20 volt upto the region where there are almost no counts. In this method the valley of decreasing counts that is generated holds a special importance as it allows us to identify the photopeak region. Actually, when measurements done in threshold mode, in the photopeak region there is a sudden drop of the counts. The reason is that when we shift the lower threshold obviously the total counts beyond the instantaneous value will gradually decrease as we are rejecting all the signal generating a voltage less than that value. But in the photopeak region most of the counts come under this so that as we move through the photopeak region the rate of decrement increases. This allows us to identify the photopeak region as that will be needed in the window mode.

Window mode

Window mode: In the window mode we basically construct a gate of voltage. Every signal generating a voltage within that gate is allowed to be considered. This gate is constructed by setting the baseline voltage to an initial value denoting the lower end of the gate and then setting the window to a certain defining the length of the gate. Now the window is kept fixed and the

baseline is shifted in steps(step length is equal to the value of the window) to record the data accordingly. So here we are taking the data in slices to see the number of counts in each slice as this is needed to construct the whole energy spectrum. We have set the window value to 0.20 volt up to the end of the Compton region and shifted the baseline in steps of 0.20 volt. For the photopeak region we have set the window value to 0.02 volt and shifted the baseline in steps of 0.02 volt. This is done because our region of interest is only the photopeak region, so that we need to record with a higher precision. After the end of photopeak region we have sufficient data to construct the whole spectrum.

Experimental Results

^{60}Co results

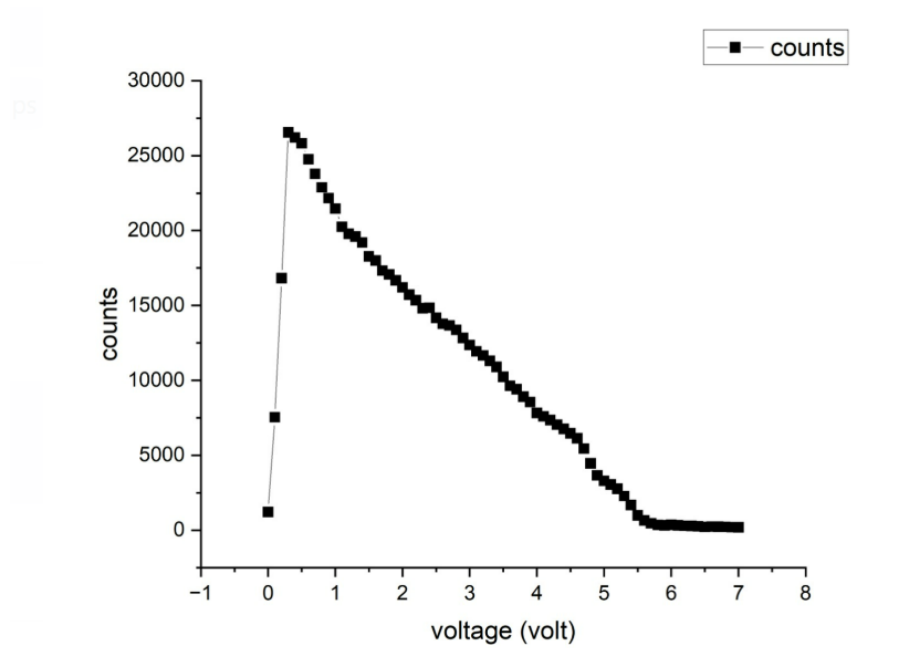


Figure 1.2: Spectrum of ^{60}Co in Threshold mode tube.

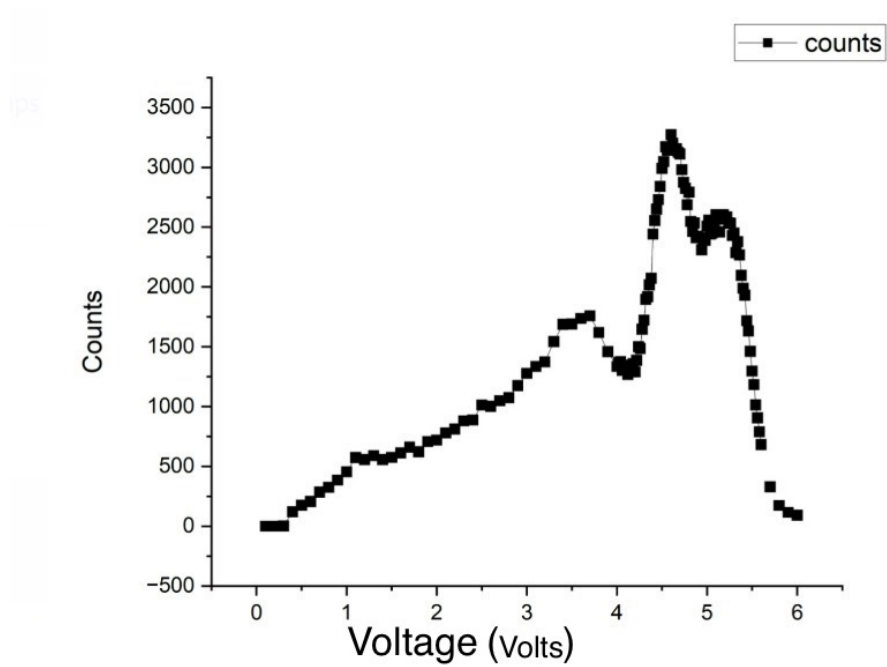


Figure 1.3: Spectrum of ^{60}Co in Window mode tube.

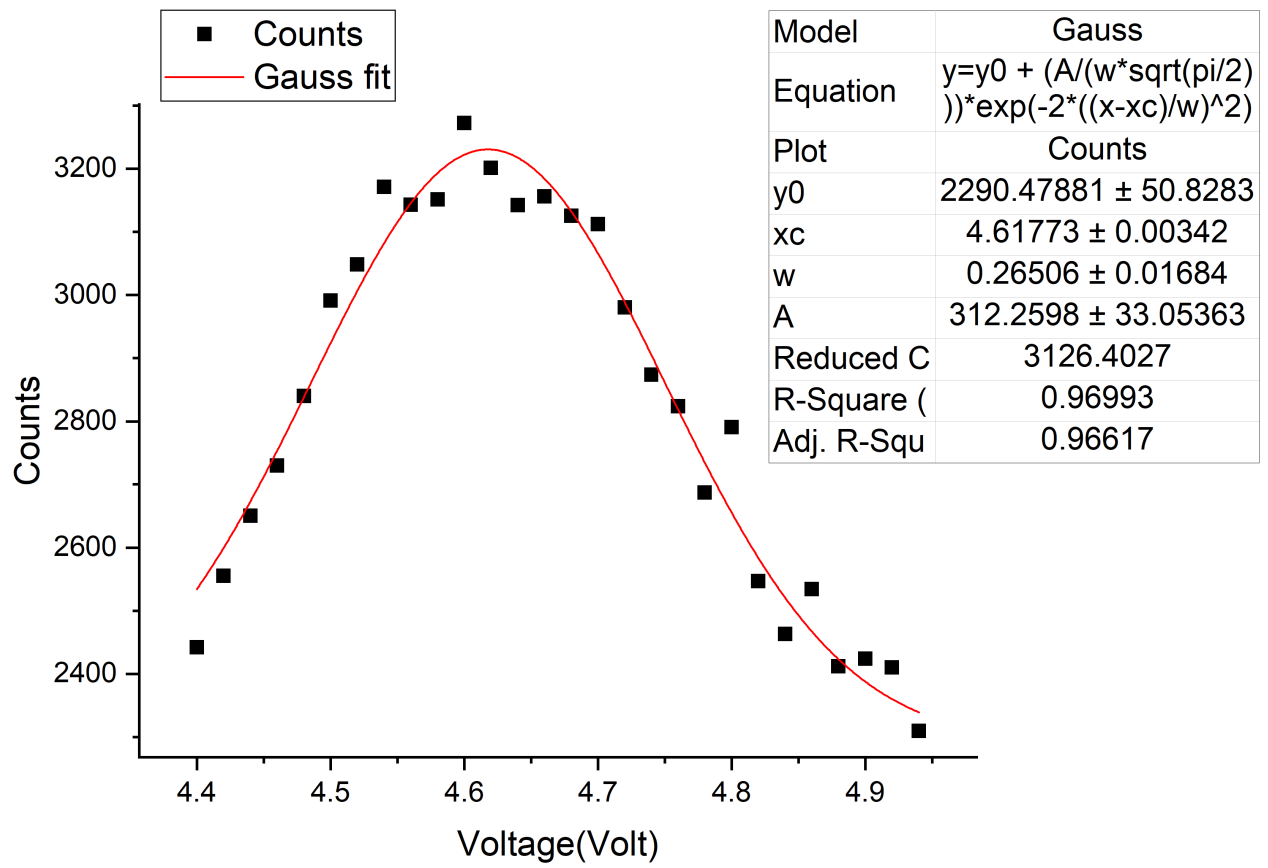


Figure 1.4: Gaussian fit of photopeak region of 1173 keV peak in the spectrum of ^{60}Co .

For 1173keV:

- The FWHM of peak is 0.310 ± 0.019 .
- The centroid of peak is 4.610 ± 0.003 .
- The energy resolution for this energy is then $(6.750 \pm 0.425)\%$.

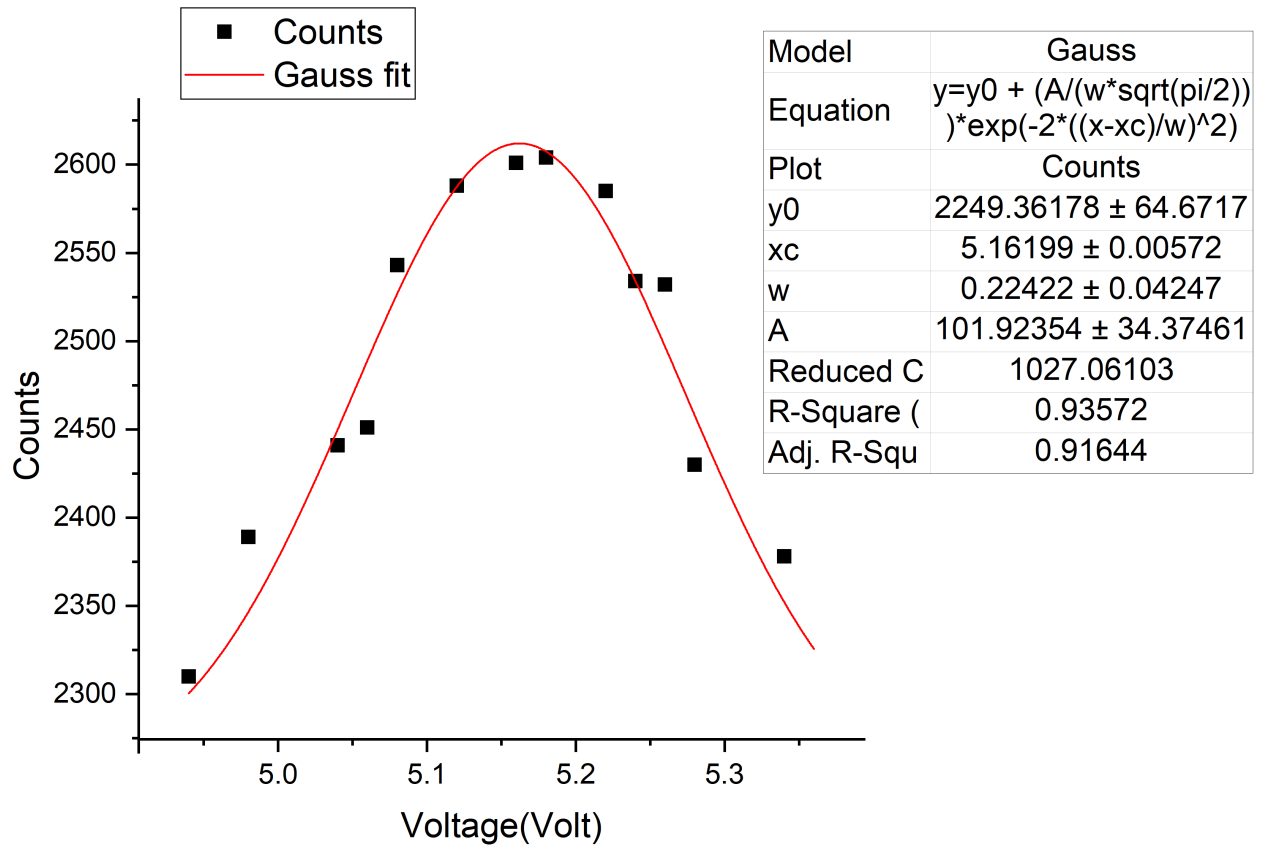


Figure 1.5: Gaussian fit of photopeak region of 1332 keV peak in the spectrum of ^{60}Co .

For 1332 keV:

- The FWHM of peak is 0.264 ± 0.050 .
- The centroid of peak is 5.160 ± 0.005 .
- The energy resolution for this energy is then $(5.110 \pm 0.965)\%$.

^{137}Cs results

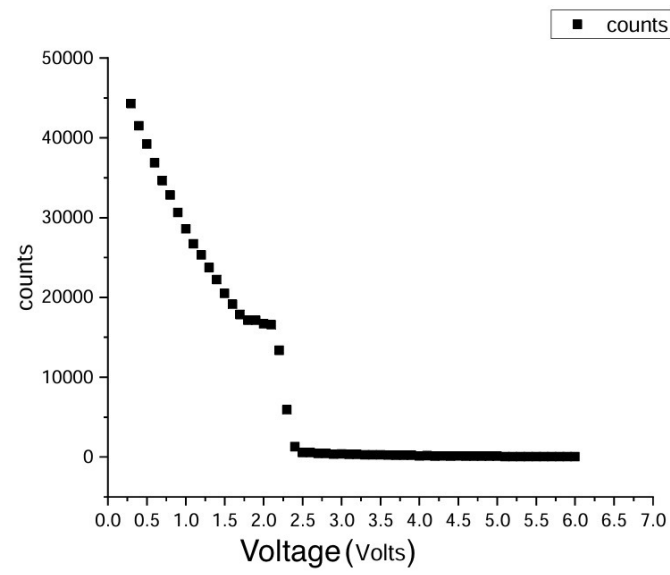


Figure 1.6: Spectrum of ^{137}Cs in Threshold mode tube.

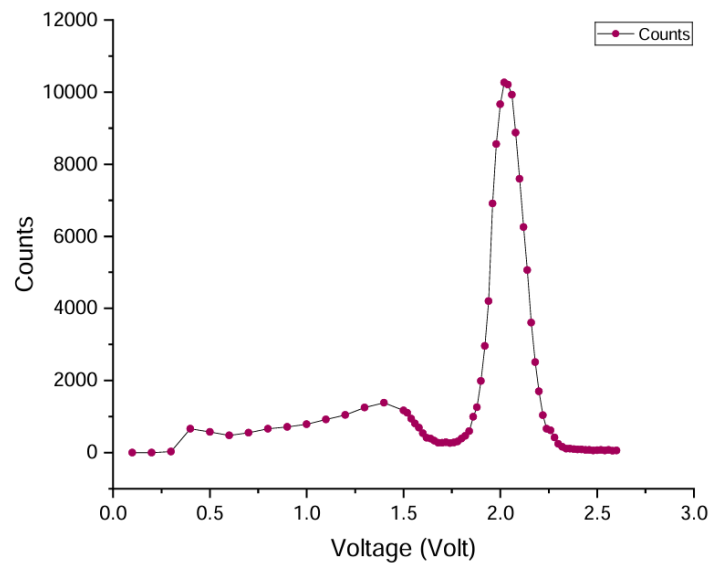


Figure 1.7: Spectrum of ^{137}Cs in Window mode tube.

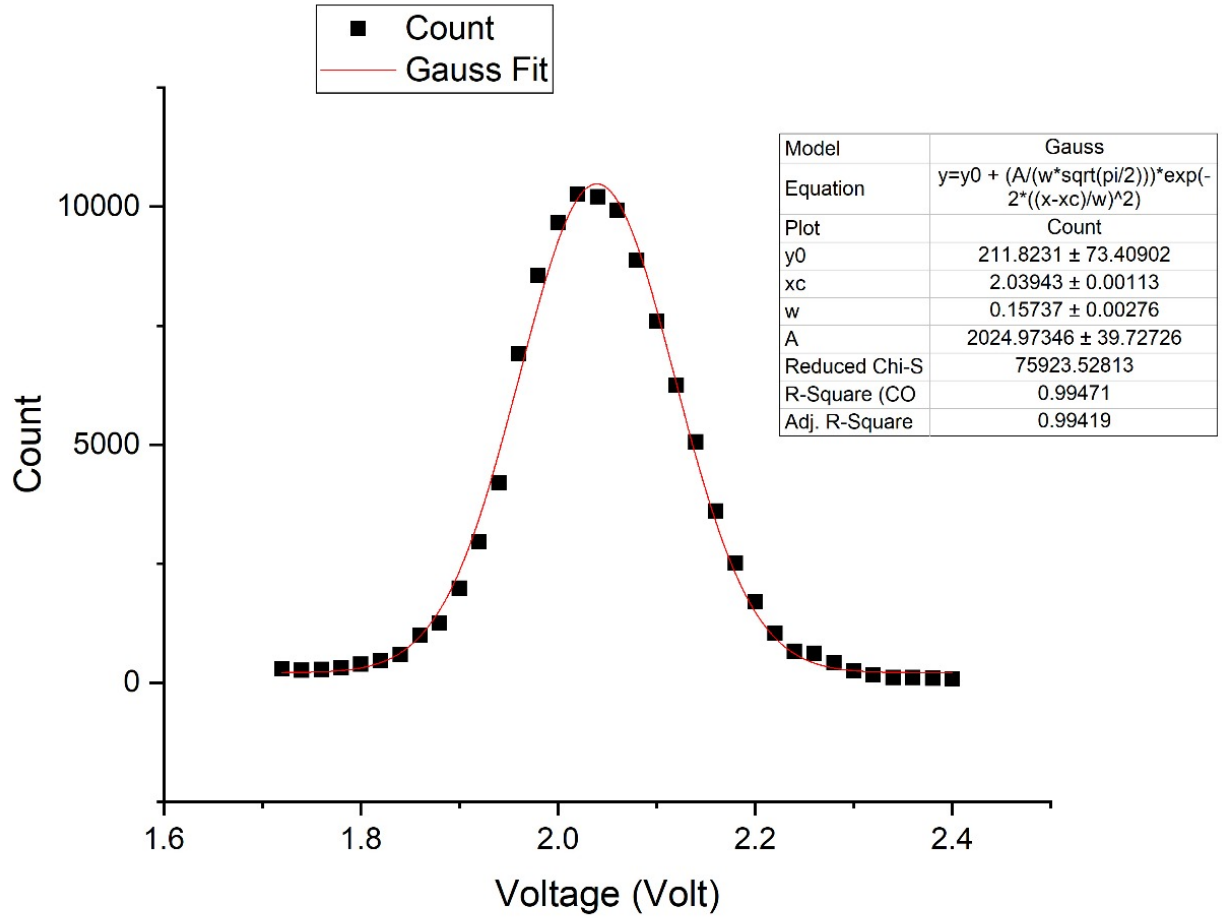


Figure 1.8: Gaussian fit of photo peak of ^{137}Cs

- The FWHM of peak is 0.185 ± 0.003
- The centroid of peak is 2.039 ± 0.001
- The energy resolution for this energy is then 9.080 ± 0.155

Energy (keV)	Resolution
662	9.080 ± 0.155
1172	6.750 ± 0.425
1332	5.110 ± 0.965

Table 1.1: Energy resolution at different gamma-ray energies.

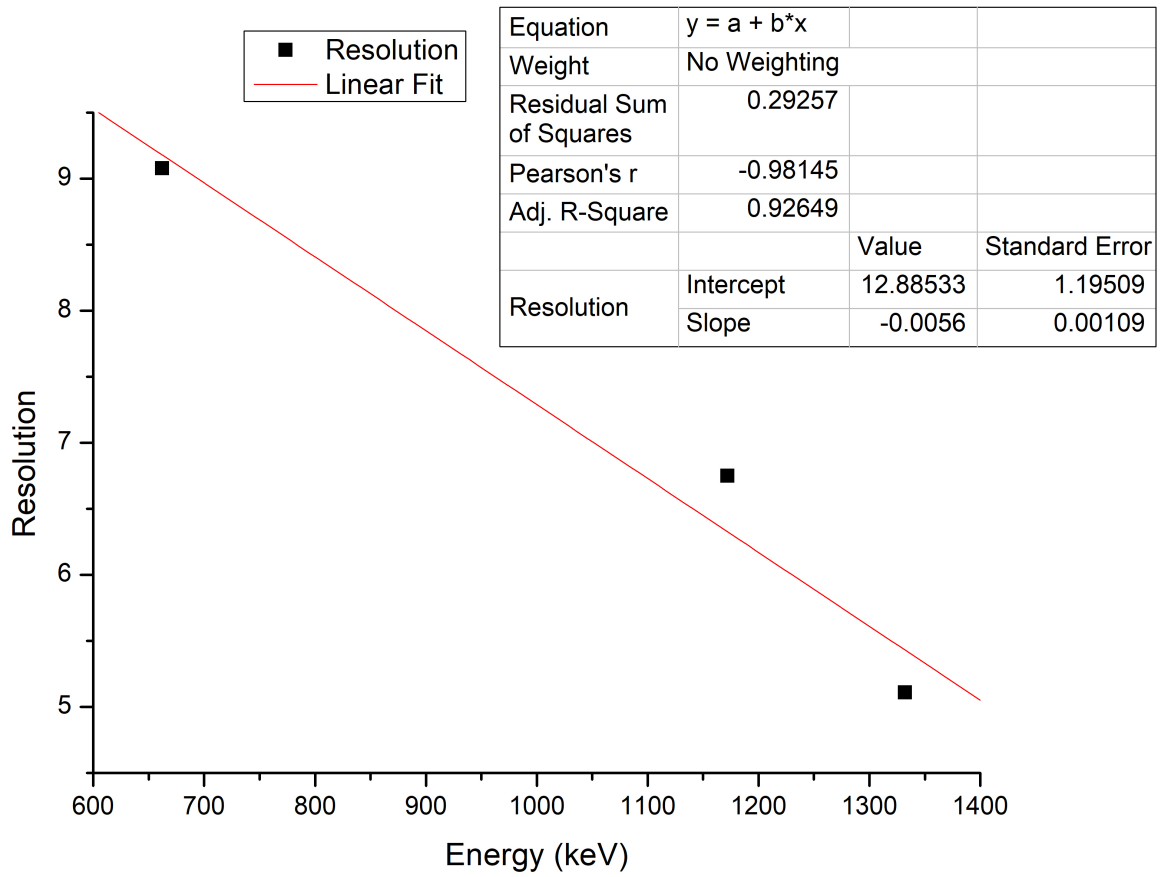


Figure 1.9: Resolution vs Energy

The nature of the calibration is as expected and the slope of this curve is -0.00524 ± 0.0011 and intercept is 12.43991 ± 1.20379 .

1.3 Discussion and Conclusion

- i) NaI detector is an inorganic scintillator detector which can emit scintillation photons within a very short interval of time which makes them very efficient in timing measurements. But in energy measurements they are not upto that mark as can be seen from our experiment. For very precise energy measurements this scintillator will not be a wise choice.
- ii) The nature of the calibration of the energy resolution vs energy curve is as expected, i.e., linearly decreasing energy resolution with increasing energy which agrees well with the theory.

Experiment 2

Detection of gamma radiation of known sources using Sodium Iodide detector and multichannel analyzer (MCA)

Objective

1. To find the energy resolution of the detector as a function of gamma energy.
2. To find the efficiency of the detector as a function of energy.
3. To plot the energy calibration of the detector and hence to find the energies of an unknown source.

2.1 Theory

Resolution and efficiency of the detector:

The detailed theory about the resolution and efficiency has been discussed in our previous experiment and so we are specifying the formulas only here.

$$Resolution = \frac{\text{FWHM of distribution}}{\text{Centroid of distribution}} \times 100\%$$

$$\text{Absolute Efficiency, } \epsilon_{abs} = \frac{\text{number of pulse recorded}}{\text{number of radiation quanta emitted by source}}$$

$$\text{Intrinsic efficiency, } \epsilon_{int} = \text{Absolute efficiency} \times \frac{4\pi}{\text{Solid angle}}$$

The relation between the absolute and intrinsic efficiency is $\epsilon_{int} = \epsilon_{abs} \times \frac{4\pi}{\Omega}$

Energy calibration

In the multichannel analyser each channel corresponds to some energy and we need to calibrate the channel v/s energy with some known sources so that we can determine the energy of any unknown radiation. We can do this by plotting the centroids of the photopeaks of the energy spectrums of the known source v/s the corresponding energies of the sources. The expected nature is linear. This linearity is the identity of a detector for energy measurements. So, by analysing the channel of occurrence of any radiation we can determine the energy of the unknown radiation.

Details of the source :- we have used the same sources here i.e., ^{60}Co , ^{137}Cs , ^{22}Na

2.1.1 Energy Callibration

The energy calibration of the detector is shown in Fig 2.1 below. The data for calibration is shown in Table 2.1 below.

Channel no	Energy (keV)
168	511
207	662
361	1173
410	1274
427	1332

Table 2.1

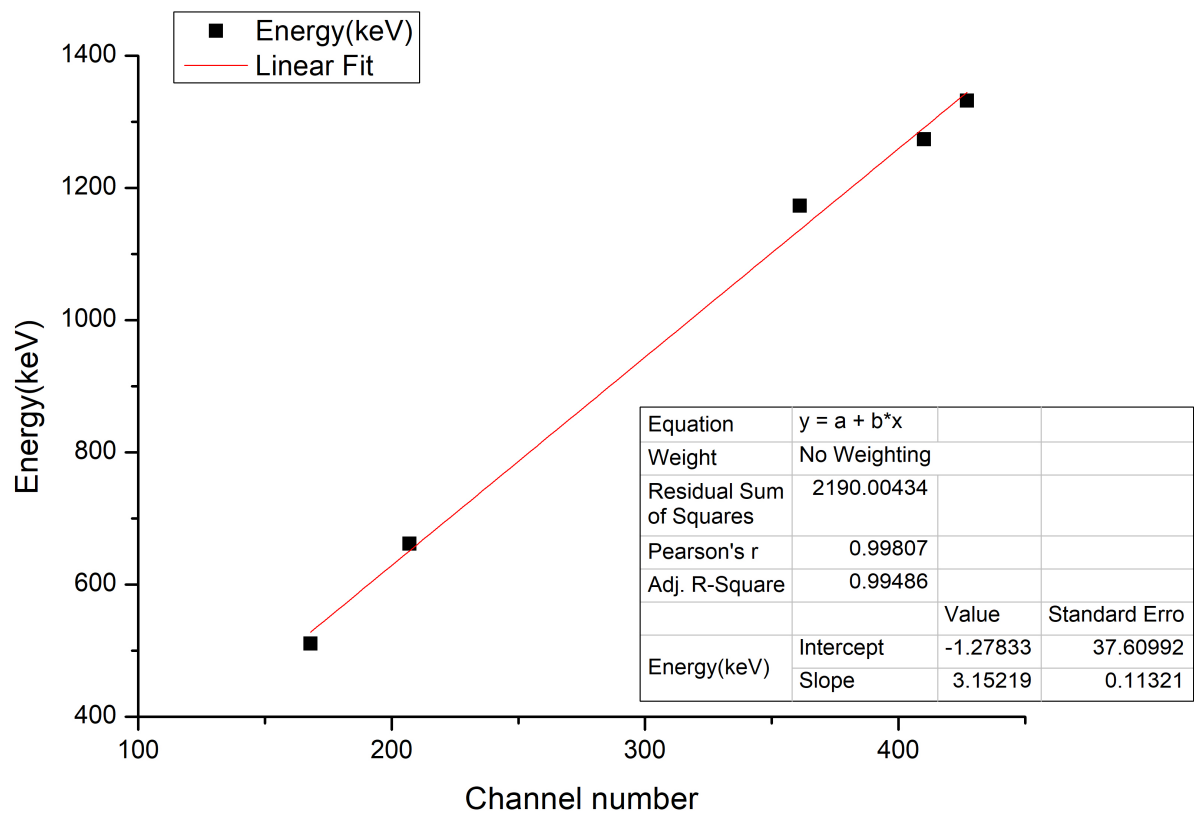


Figure 2.1: Energy Calibration of detector

2.2 Experimental Results

^{60}Co

Real time = 600 second, High voltage = 750 volts

The spectrum obtained is shown in Figure [2.2](#)

The first photo peak is at channel no. 361 and the corresponding energy is 1173 keV.

The first photo peak is at channel no. 427 and the corresponding energy is 1332 keV.

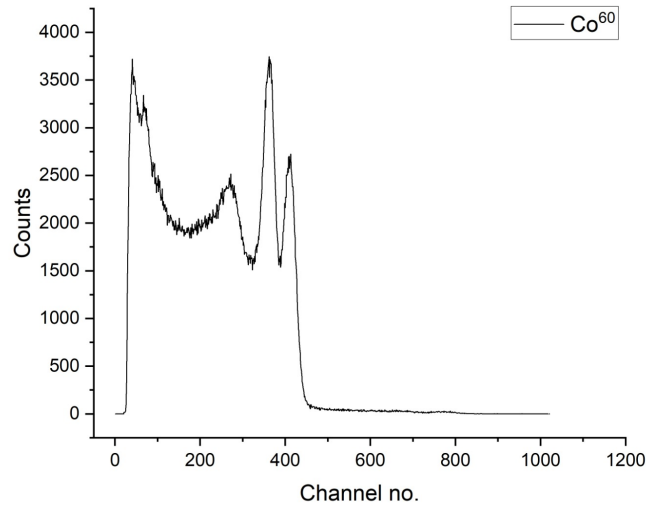


Figure 2.2: Energy spectrum of ^{60}Co

^{137}Cs

Real time = 600 s ,High voltage = 750 volts

The spectrum obtained is shown in Figure [2.3](#) below.

The first photo peak is at channel no. 207 and the corresponding energy is 662 keV.

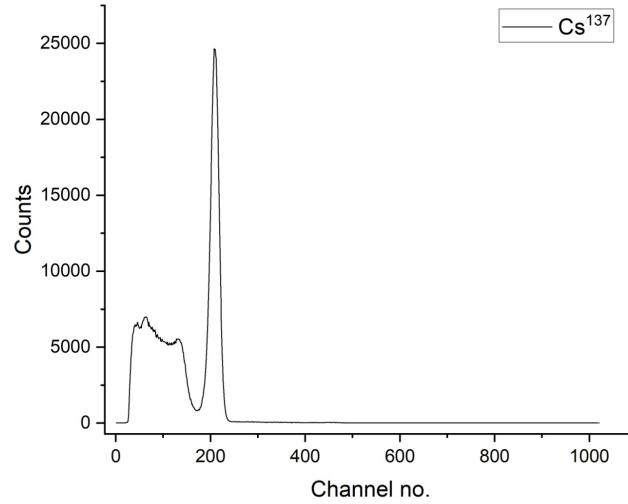


Figure 2.3: Energy spectrum of ^{137}Cs

^{22}Na

Real time = 1200 s

High voltage = 750 volts

The spectrum obtained is shown in Fig 2.4 below.

The annihilation peak is at channel no. 168 and the corresponding energy is **511 keV**.

The photo peak is at channel no. 410 and the corresponding energy is **1274 keV**.

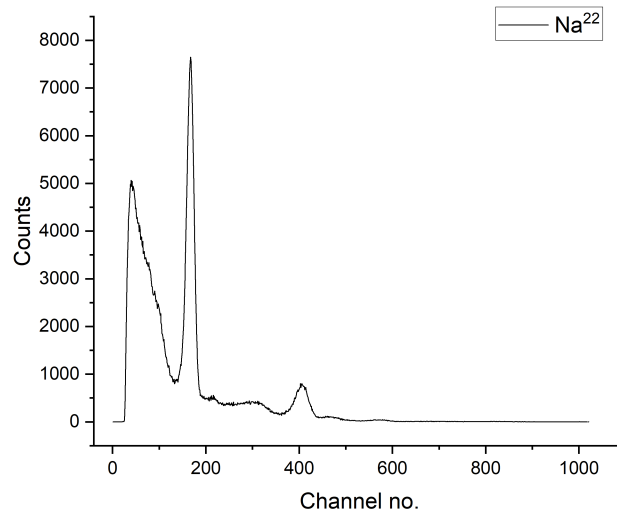


Figure 2.4: Energy Spectrum of Na^{22}

For ^{137}Cs

It has one gamma of energy 662 keV

Half-life is 30 years.

Activity on July, 2011 is 108 kBq = 108000 Bq.

Present activity is 79030 Bq.

Counts for 600 seconds is 47418000, which is the theoretical value. Now, the relative yield for this gamma is 85.1%. So, the actual counts should be 40305300.

Now, our experimental counts are 522147 ± 880 .

So, the absolute efficiency for energy 662 keV is

$$\frac{522147 \pm 880}{40305300} = 0.0130 \pm 0.0001 = (1.30 \pm 0.01)\%$$

Now, the radius of the detector is $a = 1.000 \pm 0.039$ inch, and the distance from the source is $d = 1.000 \pm 0.039$ inch.

So, the solid angle subtended by the detector at the source is

$$\Omega = 2\pi \times \left(1 - \frac{d}{\sqrt{d^2 + a^2}}\right) = 1.830 \pm 0.143$$

So, the intrinsic efficiency is

$$0.0130 \pm 0.0001 \times \frac{4\pi}{1.83 \pm 0.14} = 0.08900 \pm 0.00141 = (\mathbf{8.90} \pm \mathbf{0.14})\%$$

The FWHM of the photo peak is 20.03 ± 0.27 and the centroid is 207.56 ± 0.09 .

So, the resolution for this energy is

$$\frac{20.03 \pm 0.27}{207.56 \pm 0.09} = 0.0900 \pm 0.0007 = (\mathbf{9.00} \pm \mathbf{0.07})\%$$

For ^{60}Co

It has two gammas of energies 1173 keV and 1332 keV.

Half-life is **5.27 years**.

Activity on July 2011 is **121 kBq**

Present activity is **19000 Bq**.

Counts for 600 seconds is **11400000**, which is the theoretical value. Now, the relative yield for the 1173 keV gamma is 99.85% and that for 1332 keV gamma is 99.98%. So, the actual counts should be 11377200 and 11388600 for **1173 keV** and **1332 keV** gamma, respectively.

Now, our experimental counts are 57792 ± 983 and 48686 ± 776 for 1173 keV and 1332 keV, respectively.

So, the absolute efficiency for energy 1173 keV is

$$\frac{57792 \pm 983}{11377200} = 0.00500 \pm 0.00003 = (0.500 \pm 0.003)\%$$

And that for energy 1332 keV is

$$\frac{48686 \pm 776}{11388600} = 0.00420 \pm 0.00003 = (0.420 \pm 0.003)\%$$

Now, the radius of the detector is $a = 1.000 \pm 0.039$ inch, and the distance from the source is $d = (1.000 \pm 0.039)$ inch.

So, the solid angle subtended by the detector at the source is

$$\Omega = 2\pi \times \left(1 - \frac{d}{\sqrt{d^2 + a^2}}\right) = 1.830 \pm 0.143$$

So, the intrinsic efficiency for 1173 keV gamma is

$$0.00500 \pm 0.00003 \times \frac{4\pi}{1.830 \pm 0.143} = 0.0340 \pm 0.0003 = (3.40 \pm 0.03)\%$$

And that for 1332 keV gamma is

$$0.00420 \pm 0.00003 \times \frac{4\pi}{1.830 \pm 0.143} = 0.0280 \pm 0.0003 = (2.80 \pm 0.03)\%$$

The FWHM of the photo peak of 1173 keV is 23.21 and the centroid is 370.07.

So, the resolution for this energy is

$$\frac{23.21 \pm 0.80}{370.07 \pm 0.20} = 0.0630 \pm 0.0014 = (6.30 \pm 0.14)\%$$

The FWHM of the photo peak of 1332 keV is 23.05 ± 0.91 and the centroid is 411.07 ± 0.16 .

So, the resolution for this energy is

$$\frac{23.05 \pm 0.91}{411.07 \pm 0.16} = 0.056 \pm 0.001 = (5.60 \pm 0.10)\%$$

For ^{22}Na

It has two gammas of energies **511 keV** and **1274 keV**.

Half-life is 2.6 years.

Activity on July, 2011 is 7 kBq = 7000 Bq.

Present activity is **168** Bq.

Counts for 1200 seconds is 201600, which is the theoretical value. Now, the relative yield for the 511 keV gamma is 179.91% and that for 1274 keV gamma is 99.94%. So, the actual counts should be 360864 and 201398 for 511 keV and 1274 keV gamma, respectively.

Now, our experimental counts are 125888 ± 750 and 19319 ± 455 for 511 keV and 1274 keV, respectively.

So, the absolute efficiency for energy 511 keV is

$$\frac{125888 \pm 750}{360864} = 0.348 \pm 0.001 = (34.88 \pm 0.10)\%$$

And that for energy 1274 keV is

$$\frac{19319 \pm 455}{201398} = 0.0950 \pm 0.0009 = (9.50 \pm 0.09)\%$$

Now, the radius of the detector is $d = (1.000 \pm 0.039)$ inch, and the distance from the source is $a = (1.000 \pm 0.039)$ inch.

So, the solid angle subtended by the detector at the source is

$$\Omega = 2\pi \times \left(1 - \frac{d}{\sqrt{d^2 + a^2}}\right) = 1.830 \pm 0.143$$

So, the intrinsic efficiency for 511 keV gamma is

$$0.3488 \pm 0.001 \times \frac{4\pi}{1.830 \pm 0.143} = 2.393 \pm 0.001 = (\mathbf{239.30 \pm 0.10})\%$$

And that for 1274 keV gamma is

$$0.0950 \pm 0.0009 \times \frac{4\pi}{1.830 \pm 0.143} = 0.6520 \pm 0.0009 = (\mathbf{65.20 \pm 0.09})\%$$

Now, the FWHM of the photopeak of 511 keV is 20.2, and the centroid is 171.

So, the resolution for this energy is

$$\frac{20.20 \pm 0.25}{171.00 \pm 0.09} = 0.1180 \pm 0.0008 = (\mathbf{11.80 \pm 0.08})\%$$

The FWHM of the photopeak of 1274 keV is 32.57 ± 0.97 , and the centroid is 435.00 ± 0.19 .

So, the resolution for this energy is

$$\frac{32.57 \pm 0.97}{435.00 \pm 0.19} = 0.0750 \pm 0.0012 = (\mathbf{7.50 \pm 0.12})\%$$

The calibration of resolution vs energy is shown in Fig 2.5.

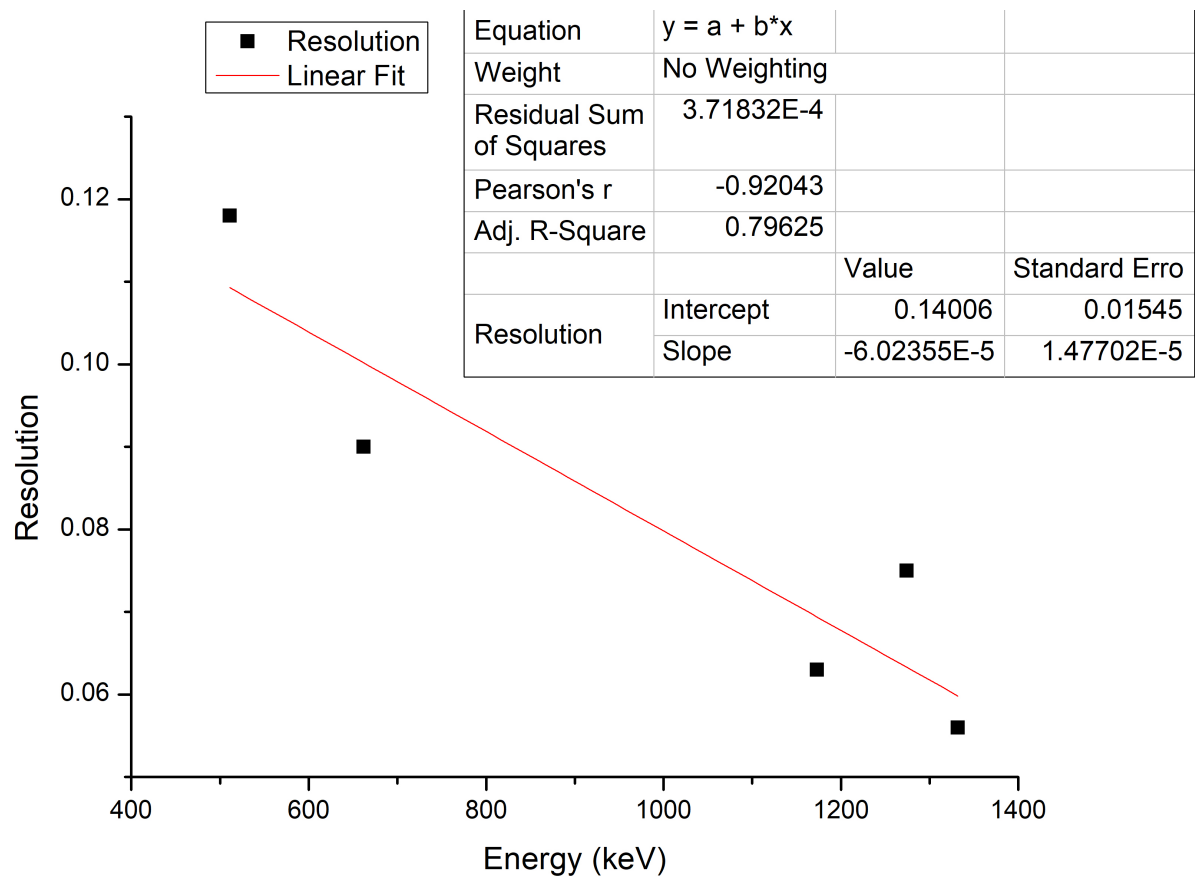


Figure 2.5: Resolution Vs Energy(keV) Graph

The plot of efficiency calibration is shown in Fig 2.6.

Energy (keV)	Intrinsic efficiency (%)
511	239.30 ± 0.01
662	8.90 ± 0.14
1172	3.40 ± 0.03
1274	65.20 ± 0.09
1332	2.80 ± 0.03

Table 2.2

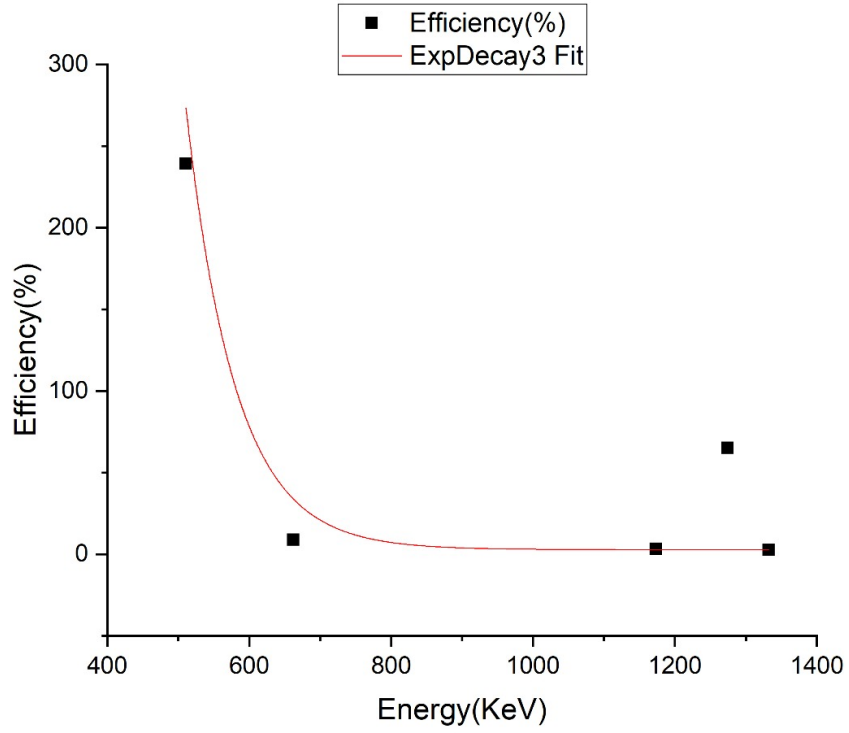


Figure 2.6: Intrinsic efficiency vs Energy

The calibration of intrinsic efficiency vs energy (modified) is shown in Fig 2.7.

Energy (keV)	modified Intrinsic efficiency (%)
511	28.80 ± 0.10
662	8.90 ± 0.14
1172	3.40 ± 0.03
1274	2.50 ± 0.03
1332	2.80 ± 0.03

Table 2.3

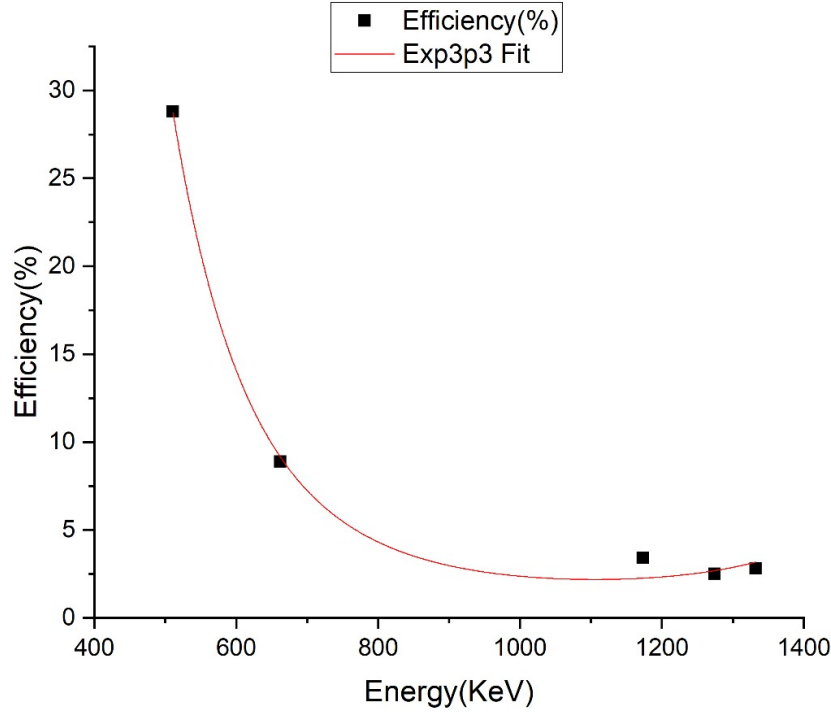


Figure 2.7: Intrinsic efficiency vs Energy (Modified)

2.3 Discussion and Conclusion

1. The nature of the resolution calibration is quite linear as expected.
2. The efficiency should also decrease gradually with increasing energy. But as we see from the above table, the efficiencies of ^{137}Cs (662) and ^{22}Na (511) are not close to each other, whereas their energies are close to each other. Also, the efficiency for the 1274 keV gamma is higher than that of 1173 keV gamma, which is quite absurd.
3. Now, from the formula of intrinsic efficiency, it depends on the solid angle and absolute efficiency. Our solid angle is fixed for the detector, so our absolute efficiency should change. And in absolute efficiency, our experimental count does not change, so our theoretical counts should change.
4. Now, if we estimate the activity of ^{22}Na from the intrinsic efficiency vs energy graph, we find that for energy 1274 keV, the intrinsic efficiency should be $(2.50 \pm 0.03)\%$, and the theoretical counts should be 5366388, and the present activity should be **4476 Bq**.
5. Considering this activity of ^{22}Na , the theoretical counts for the 511 keV gammas should be 9614448. So, the absolute efficiency should be 0.042, and the intrinsic efficiency should be $(28.8 \pm 0.1)\%$.
6. We have used these values to plot the efficiency of the detector, which is shown in Fig 2.7.

7. We see that the nature of this curve is quite better than the previous one.

2.4 Effect of radiation shielding on the previous experiment

We have repeated the previous experiment with cubical lead shield for the monochromatic source ^{137}Cs and Background radiation. The background spectrum with and without the lead shield is shown in Figure.2.8.

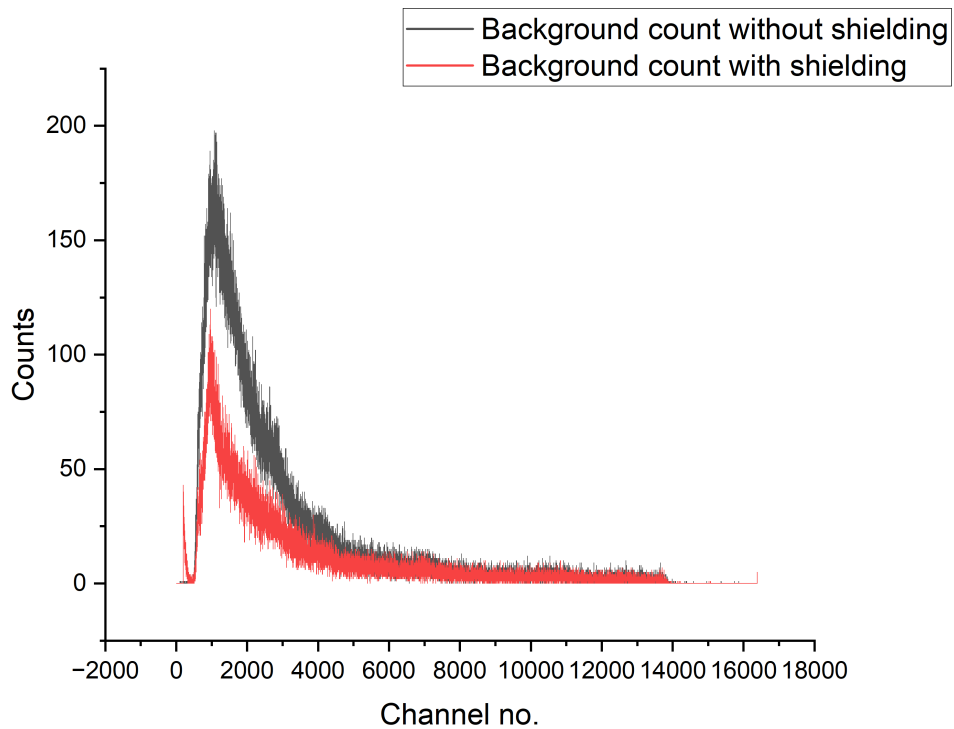


Figure 2.8: Comparision of background, with shielding and without shielding

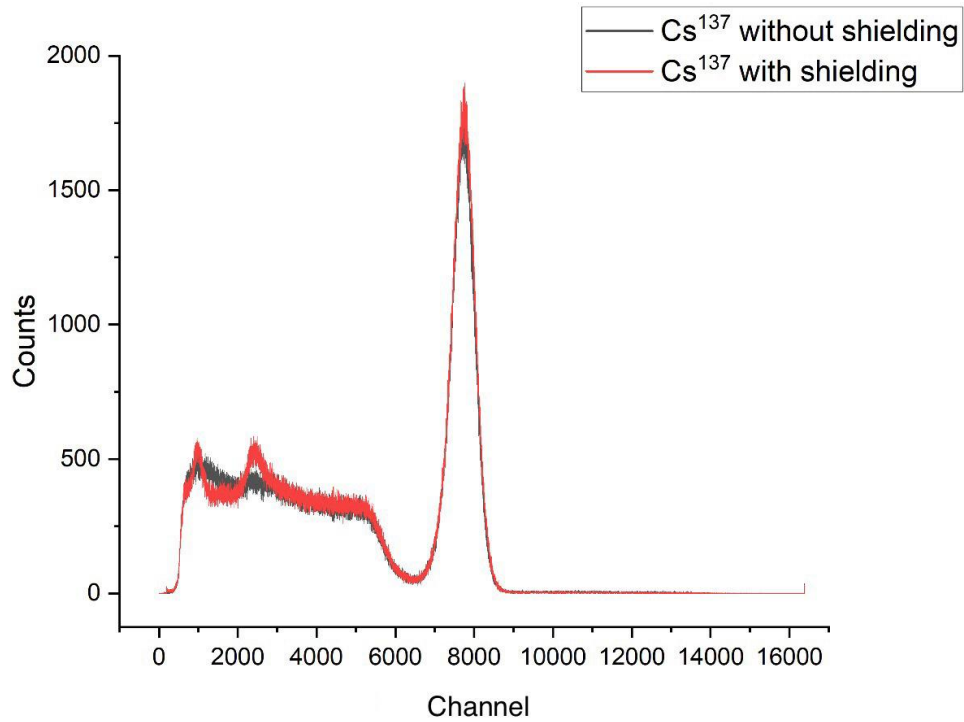


Figure 2.9: Comparison of spectrum, with shielding and without shielding

From overlapping of two spectrum we found that the count with shielding is comparatively less than the count without shielding. The shielding placed around a detector to reduce background from cosmic rays or terrestrial radiation may itself introduce a significant background from its inherent low-level radioactivity. Some additional interfering radiation can be observed as a result of the interaction of primary gamma rays from the source with shielding materials. This interaction results into peaks of Compton backscattering, X-rays(pair-production).

2.5 Discussion

1. We see that the photopeak is shifted after adding the shielding. This reason isn't yet understood by us.
2. We can also see that the background counts are suppressed by a higher amount in the low energy region only. In the higher energy region, it is not suppressed much.
3. We can also see that the channels corresponding to 74 keV to 78 keV are showing comparatively higher counts in the shielded case creating a peak also. Now, the typical X-ray energy of lead is around 75 keV. So, we may say that this peak is due to the X-rays emitted from the lead shield.
4. The noise is reduced after adding the shielding.
5. Also, we can see that the backscatter peak is enhanced after adding the shielding. This may be due to the fact that after adding the lead shield we also add a lead cap which is at 180 degrees with respect to the source and detector. So, the backscattered photons from the detector can be scattered again by the lead cap which can enhance the backscatter peak. To reduce this, we may increase the distance between the lead shield and the detector.
6. To reduce the X-rays, we may use graded shielding instead of using only lead shield.

Experiment 3

Detection of Gamma Radiation and Analysis of Angular Correlation

3.1 Objectives

1. To study the angular correlation of gamma rays emitted from ^{22}Na source in coincidence mode.

3.2 Theory

In singles mode, each detector records gamma events independently, without requiring time correlation. For the ^{22}Na source, both the 511 keV annihilation photons and the 1274 keV de-excitation photon are detected. However, unlike coincidence mode, the setup does not distinguish whether the detected photons are part of a correlated pair or background radiation.

As a result, the angular distribution of counts is expected to be relatively uniform. Deviations from this uniformity can arise due to variations in detector alignment, source-detector distance, or environmental scattering. While singles mode lacks directional sensitivity, it provides a complete picture of the total gamma yield.

For the ^{22}Na source, an annihilation peak at 511 keV and a photopeak at 1274 keV exist. In coincidence measurements, the 511 keV gamma is selected since each annihilation event produces two photons in opposite directions. By applying the coincidence window, one detector is fixed to detect one of the 511 keV gammas, while the other detector is rotated to detect the other 511 keV gamma. This setup allows for the study of angular correlation between the two gamma rays.

3.3 Experimental Data

3.3.1 ^{22}Na Source: Coincidence Mode

The energy spectra for five different angles in coincidence mode are shown in Figures below .

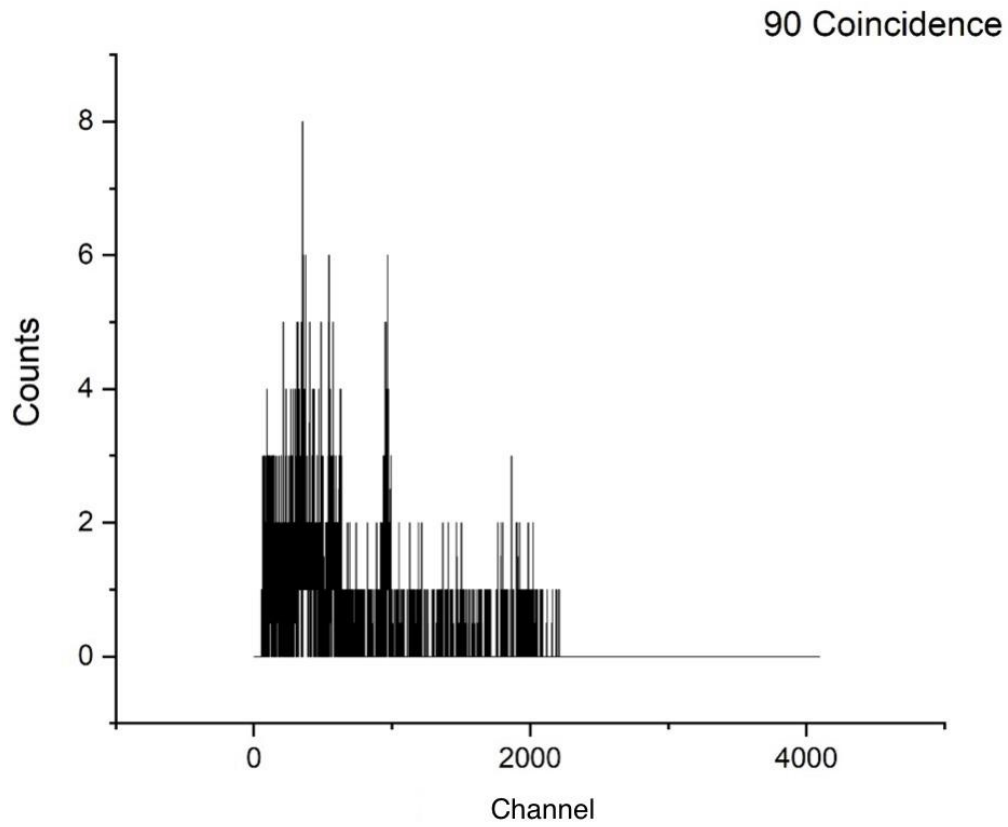


Figure 3.1: ^{22}Na spectrum at 90° in coincidence mode

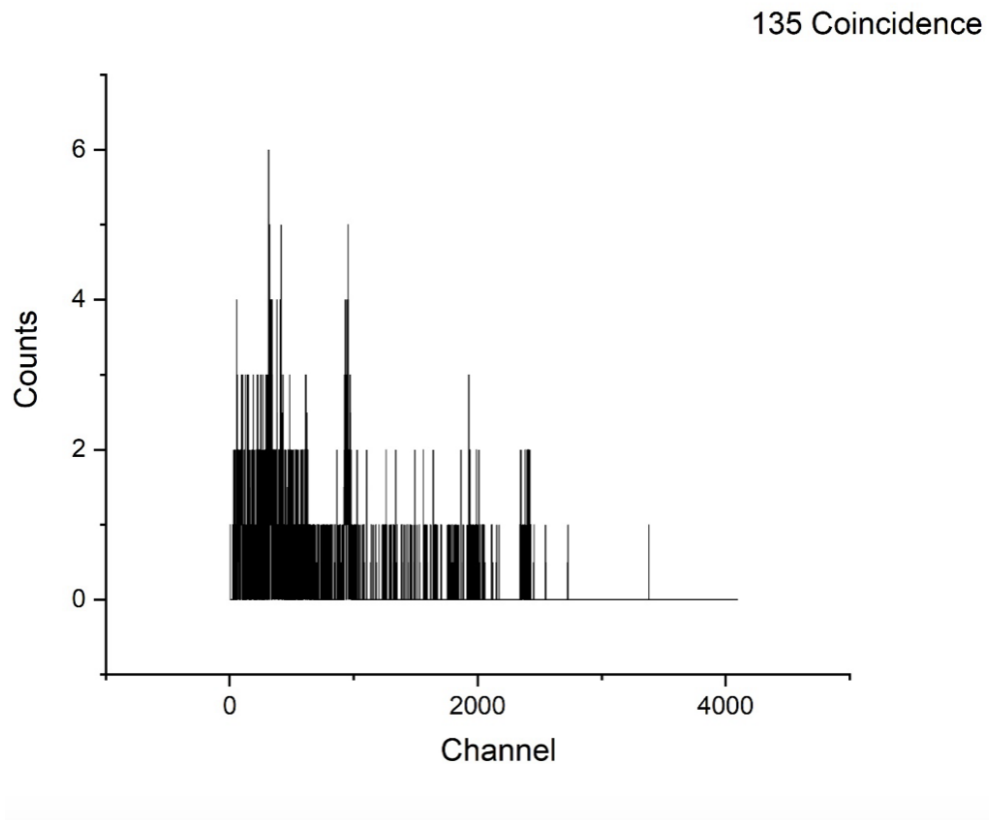


Figure 3.2: ^{22}Na spectrum at 135° in coincidence mode

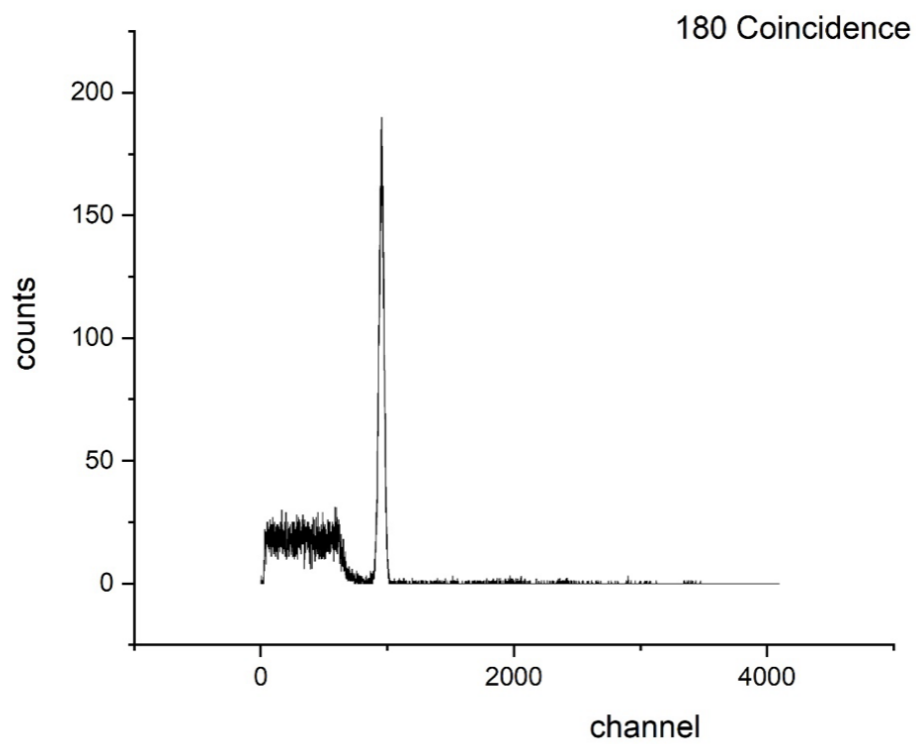


Figure 3.3: ^{22}Na spectrum at 180° in coincidence mode

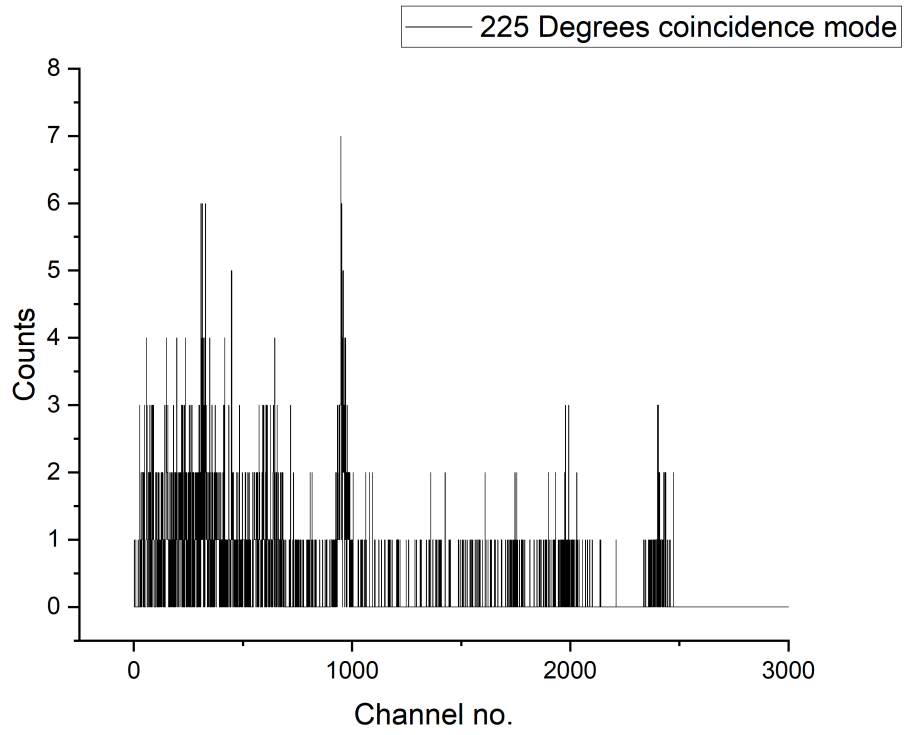


Figure 3.4: ^{22}Na spectrum at 225° in coincidence mode

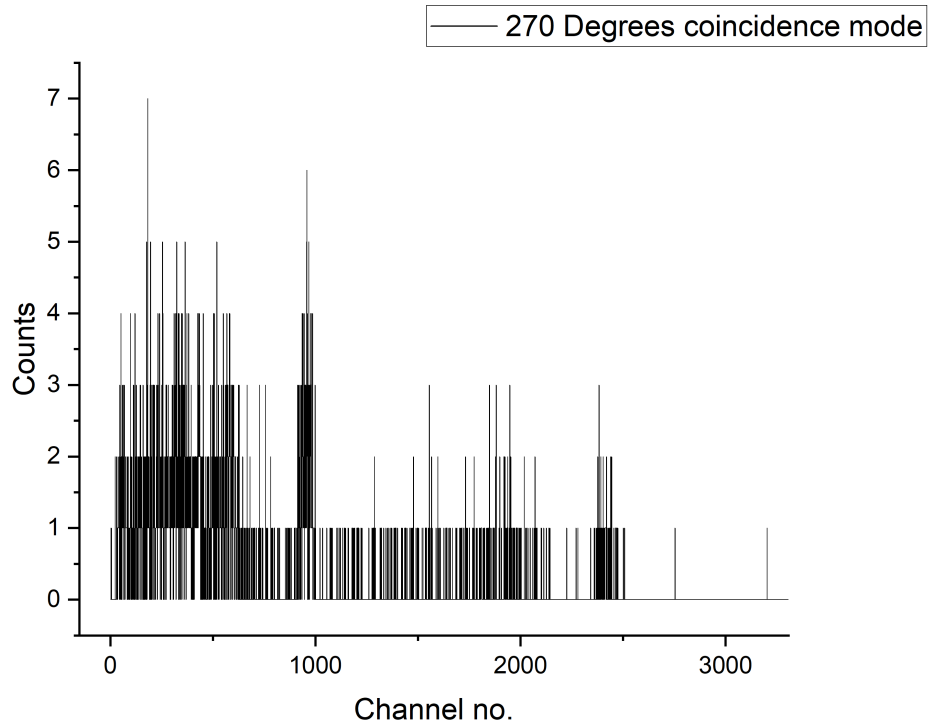


Figure 3.5: ^{22}Na spectrum at 270° in coincidence mode

3.3.2 ^{22}Na Source: Singles Mode

The energy spectra for five different angles in singles mode are shown in Figure below.

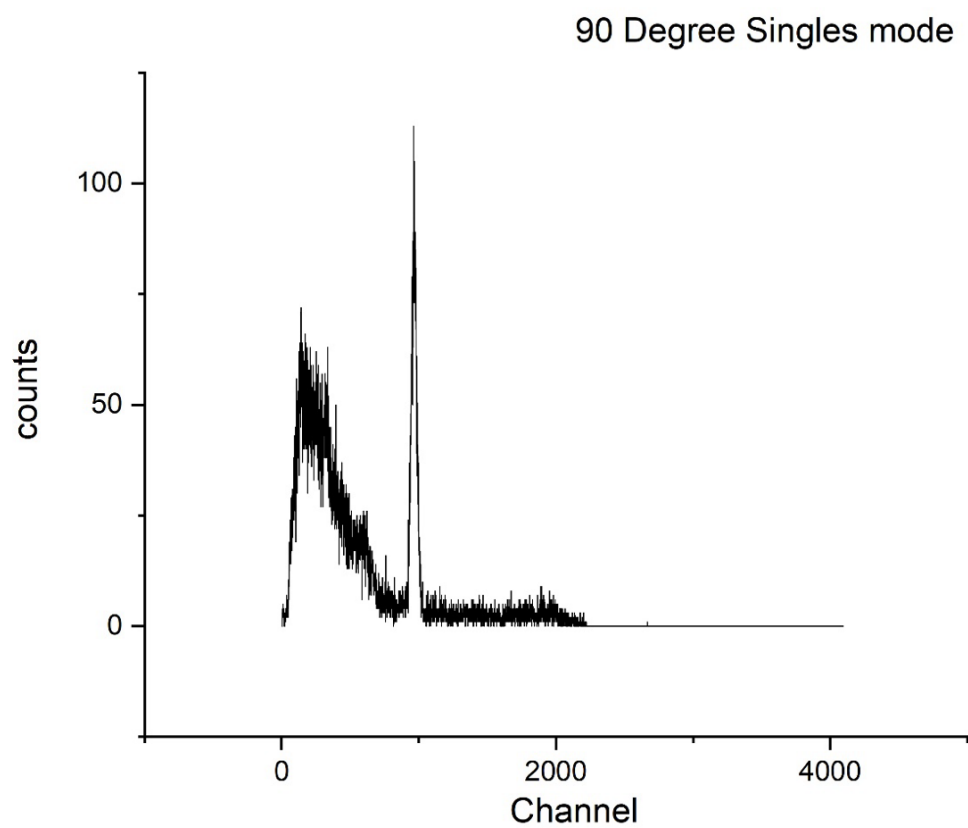


Figure 3.6: ^{22}Na spectrum at 90° in Singles mode

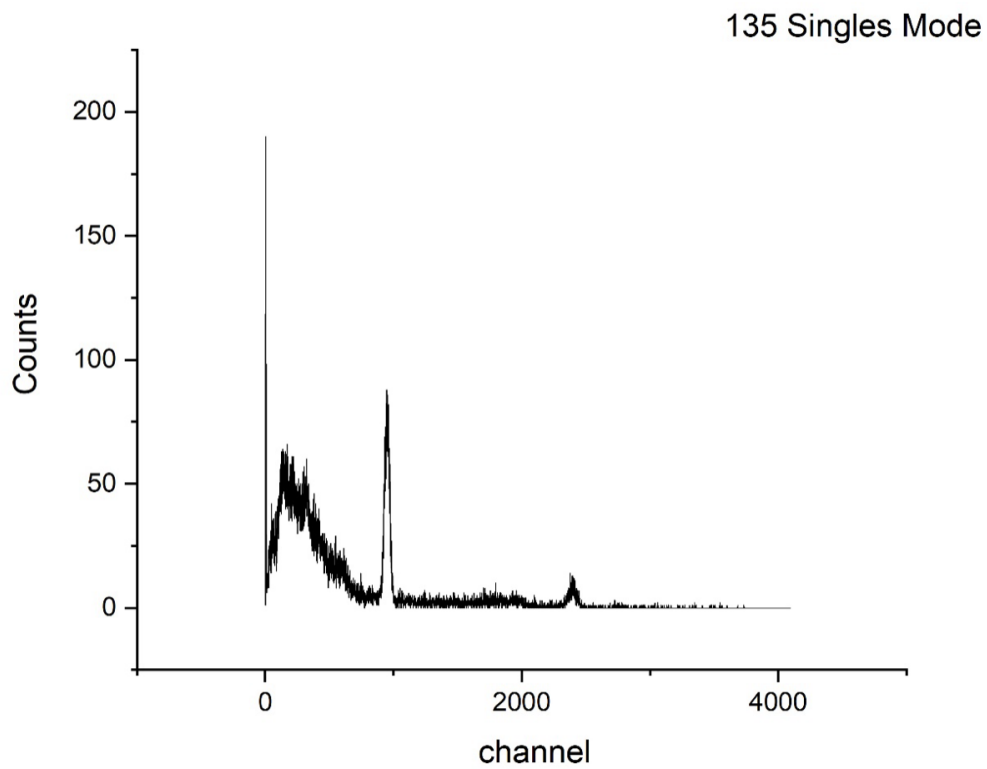


Figure 3.7: ^{22}Na spectrum at 135° in Singles mode

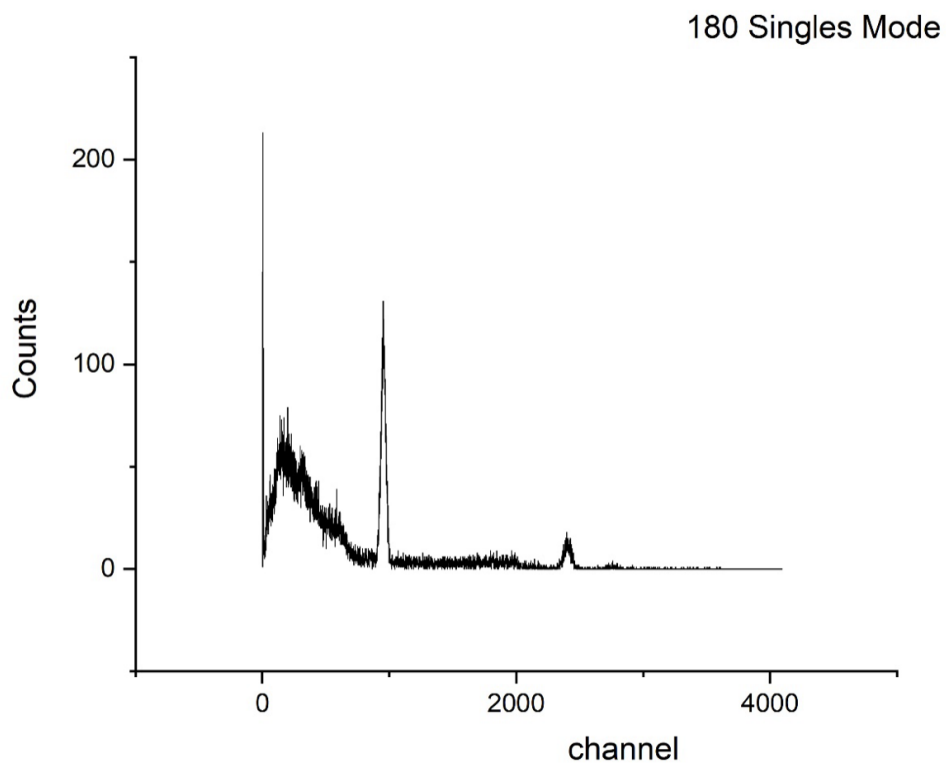


Figure 3.8: ^{22}Na spectrum at 180° in Singles mode

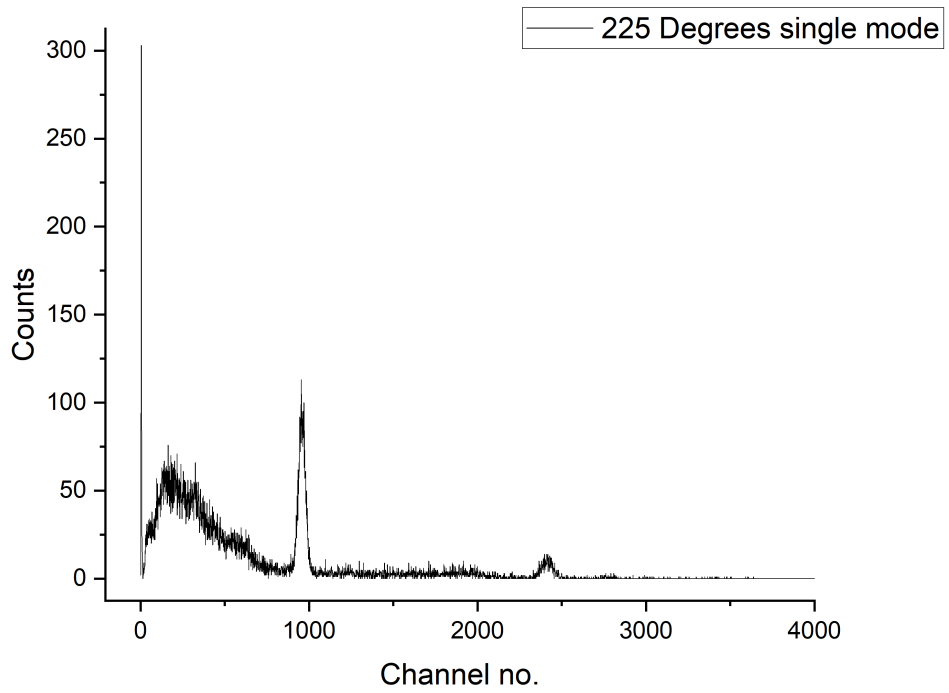


Figure 3.9: ^{22}Na spectrum at 225° in Singles mode

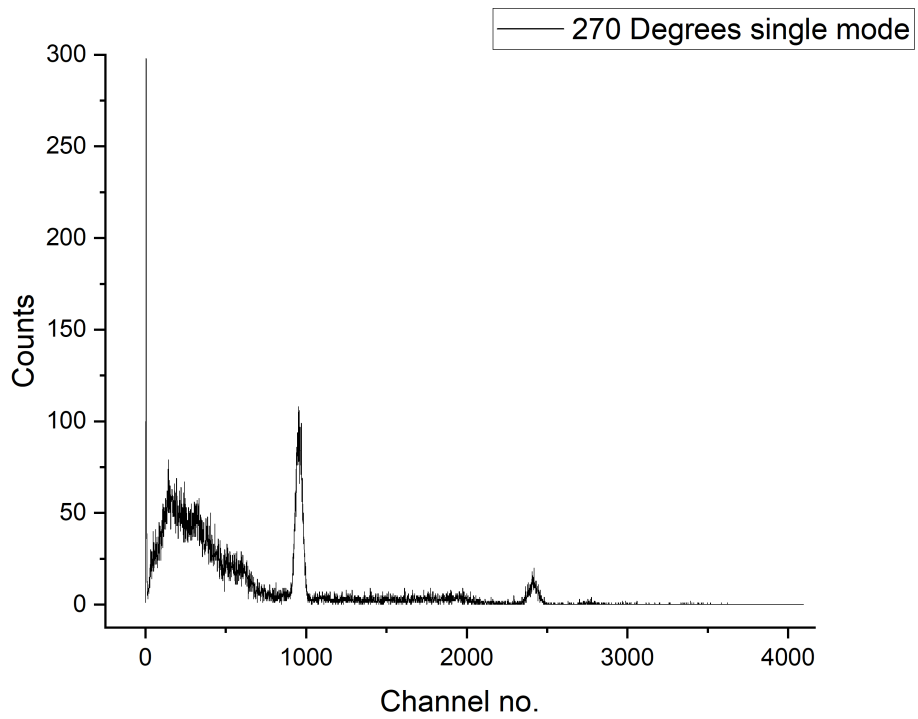


Figure 3.10: ^{22}Na spectrum at 270° in Singles mode

The photopeak counts for the 511 keV peak were recorded for different angles in both coincidence and singles mode.

Table 3.1: Photopeak Counts for 511 keV Peak in Coincidence and Singles Mode

Angle (degrees)	Coincidence Mode Counts	% Error	Singles Mode Counts	% Error
90	65	19.99	4136	2.11
135	59	21.00	3344	2.74
180	8533	1.11	5036	1.75
225	87	15.54	4397	2.25
270	37	32.78	4602	1.76

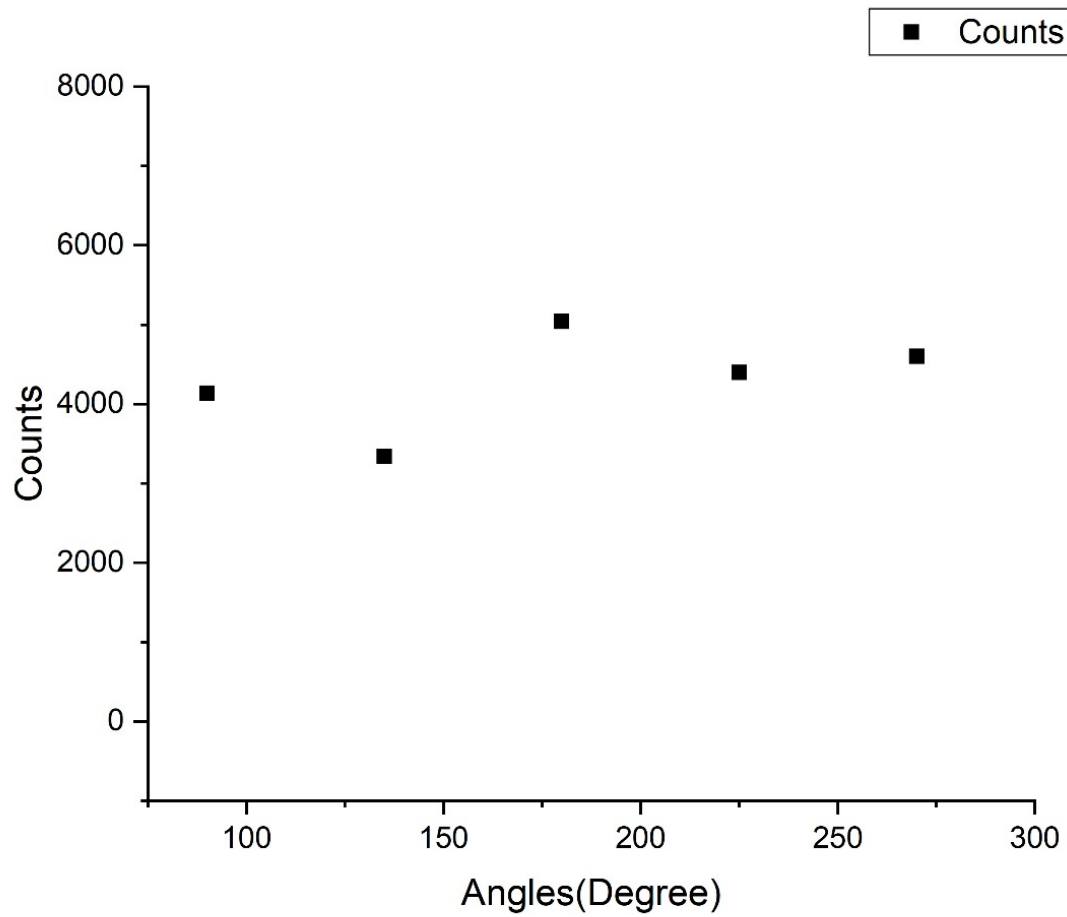


Figure 3.11: Angular distribution of Singles Modes

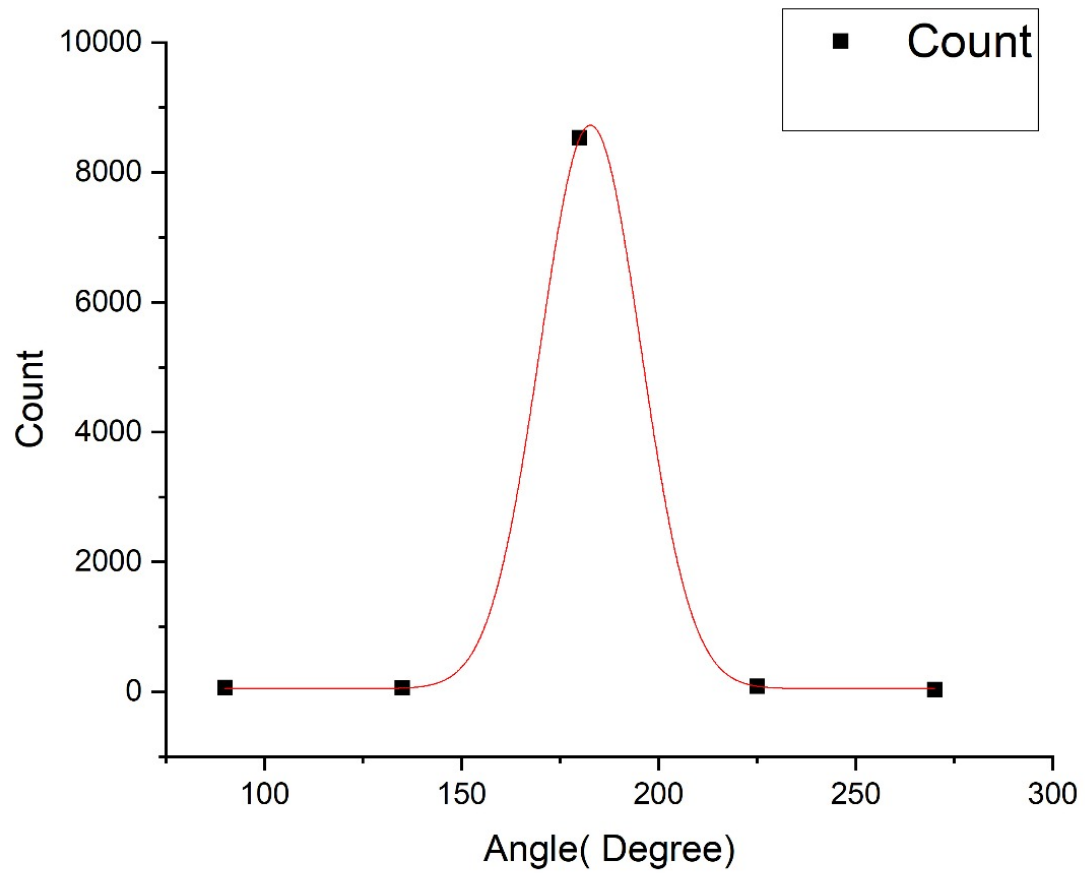


Figure 3.12: Angular distribution of coincidence Modes

3.4 Discussion

3.4.1 Expected Trends in Coincidence and Singles Mode

In coincidence mode, where only time-correlated gamma pairs were detected, we expected a clear peak at 180° , since the two 511 keV annihilation photons should be emitted exactly opposite to each other. In singles mode, where all detected gamma rays were recorded without time filtering, the counts were expected to be uniform across different angles, as the detectors should register gamma rays from all directions equally.

3.4.2 Observed Trends and Deviations

In coincidence mode, the peak at 180° was clearly visible, confirming that the annihilation gamma rays are indeed emitted in opposite directions. In singles mode, however, instead of a smooth uniform trend, the graph showed irregular fluctuations across different angles.

3.4.3 Possible Causes of the Non-Uniform Trend in Singles Mode

Variation in Source-Detector Distance: Since the setup was adjusted manually for each angle, the distance between the source and detector may not have been exactly the same each time. Small variations in distance would change the solid angle subtended by the detector, slightly affecting the count rates.

Detector Alignment Issues: If the detector position was not perfectly aligned after each movement, it could have affected the effective detection area, introducing inconsistencies in recorded counts.

Statistical Fluctuations: Data was collected at only five angles (90° , 135° , 180° , 225° , 270°), which might not be enough to completely smooth out random variations in gamma detection. A larger number of data points could have helped in identifying whether the fluctuations were due to statistical noise or a systematic effect.

Background Radiation and Scattering Effects: In singles mode, the detector does not filter out uncorrelated gamma rays, meaning it also records background radiation. Some photons might have undergone Compton scattering off nearby objects, leading to unexpected variations in the recorded counts at different angles.

Experiment 4

Coincidence study of a mixed source of Co^{60} and Ba^{133} using $CeBr_3$ detectors

4.1 Objective:

1. To calibrate the setup using known sources.
2. To study the coincidence spectra of the mixed source.

4.2 Theory

We have actually used two sources Co^{60} and Ba^{133} which we have attached with each other to analyze their superposed spectra using coincidence technique. This technique allows us to detect each gamma having a time correlation with one or many other gammas. When gamma transitions take place, there may be correlated or uncorrelated gammas in the decay scheme. If we take the data in coincidence mode then it will allow us to eliminate all the uncorrelated gammas in the decay scheme of the sources.

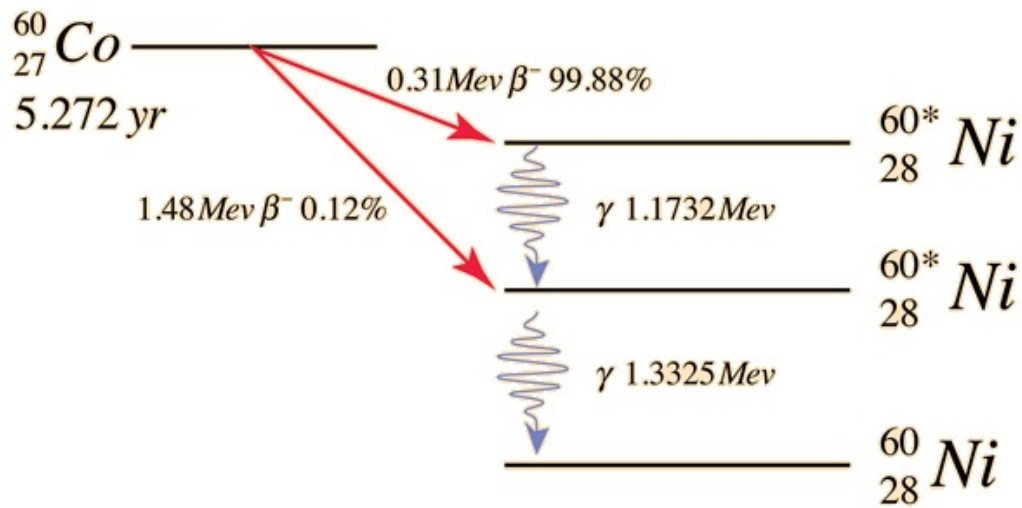


Figure 4.1: Level Scheme of Co^{60}

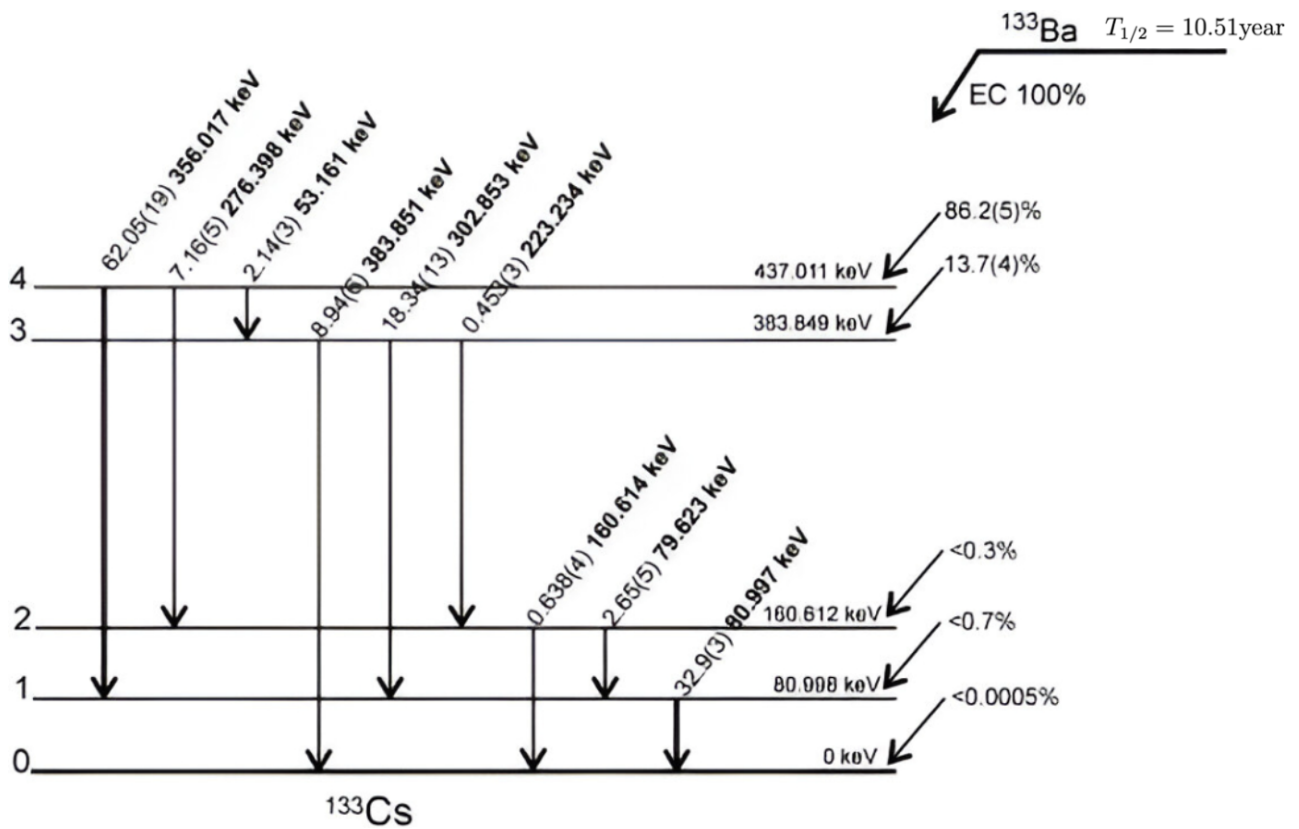


Figure 4.2: Decay Scheme of Ba^{133}

4.3 Experimental Data

The channel energy calibration is done first.

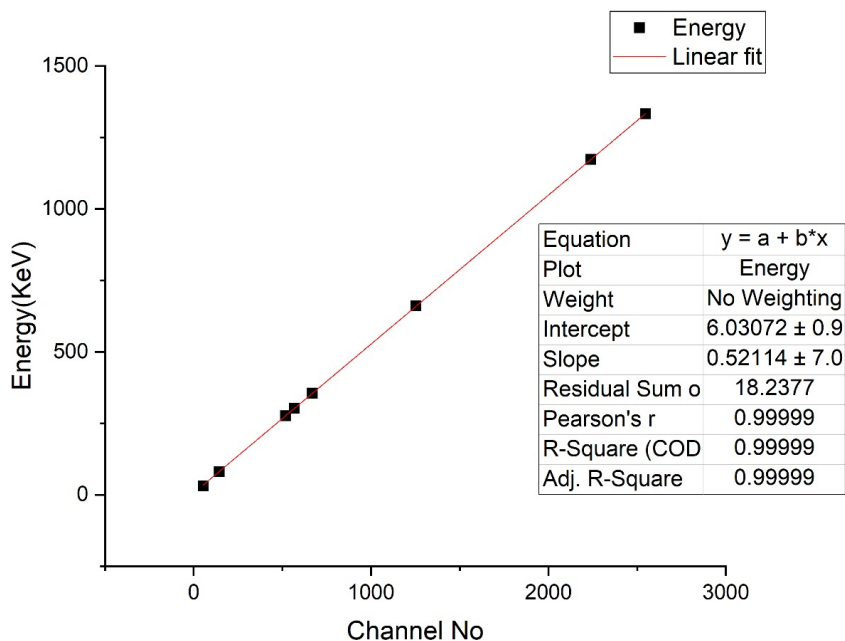


Figure 4.3: Energy calibration

The nature of the calibration is as expected and the slope of this curve is 0.52 ± 7.03 and intercept is 6.03 ± 0.93 keV.

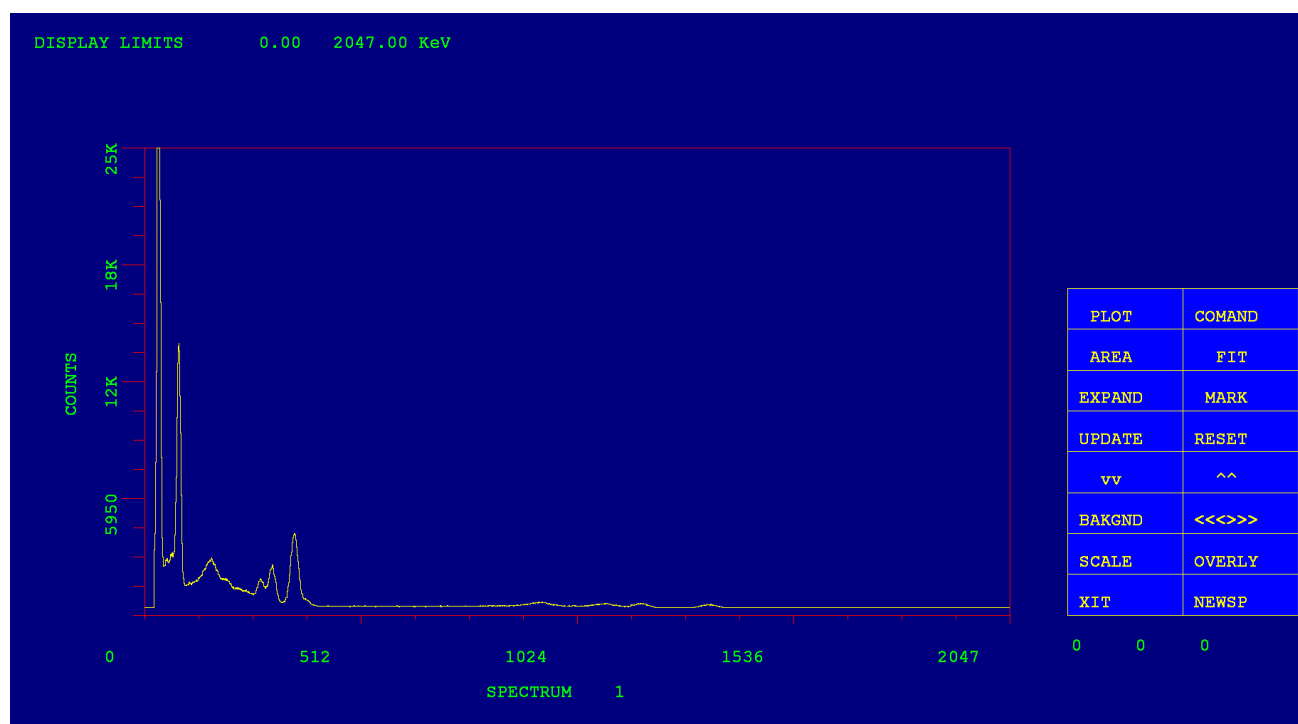


Figure 4.4: All projected spectrum of the mixed source

We have used ingasort to find the coincidence relationship for each gamma.

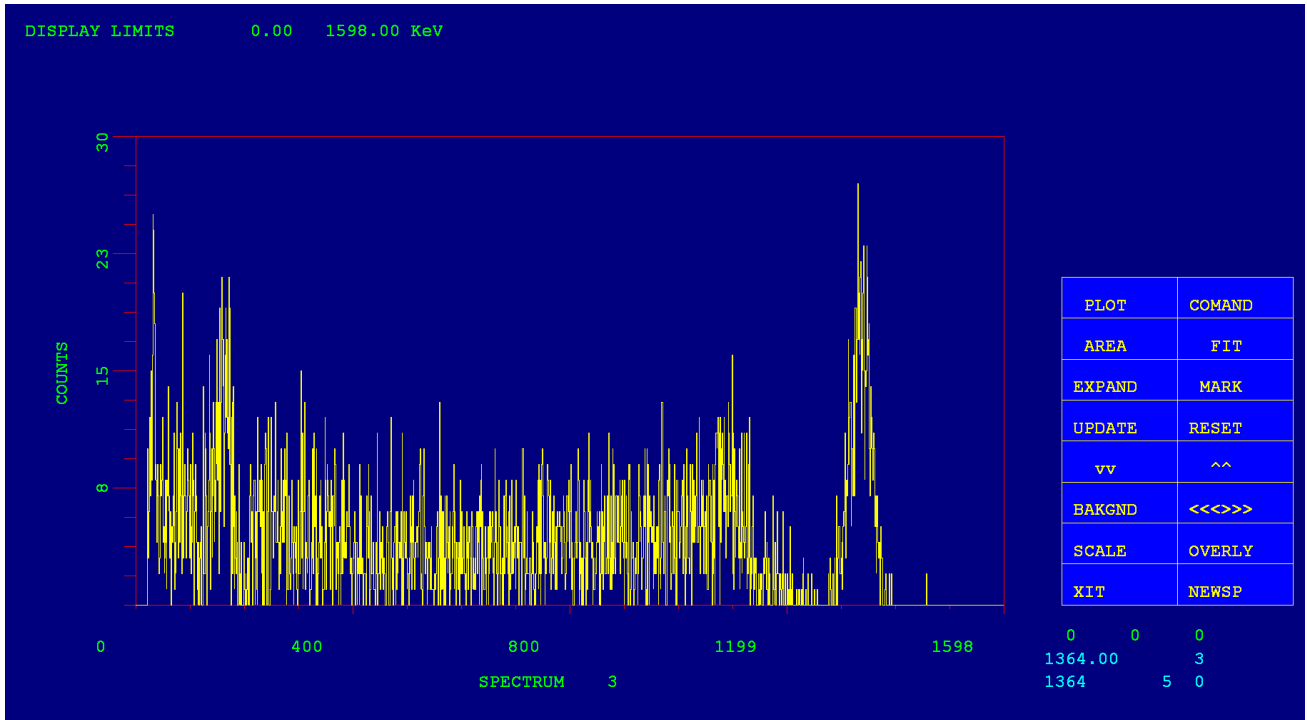
- (i) 1332 keV is in coincidence with 1173 keV.
- (ii) 1173 keV is in coincidence with 1132 keV.

Reason In the source we have Co^{60} which undergoes a beta decay and Ni^{60} is produced in an excited state which de-excites to the ground state via a two-step gamma emission; first 1173 keV gamma is emitted and then 1332 keV is emitted consecutively, which are correlated in time with each other. Since this is the most probable decay mechanism, we have detected this one in our system.

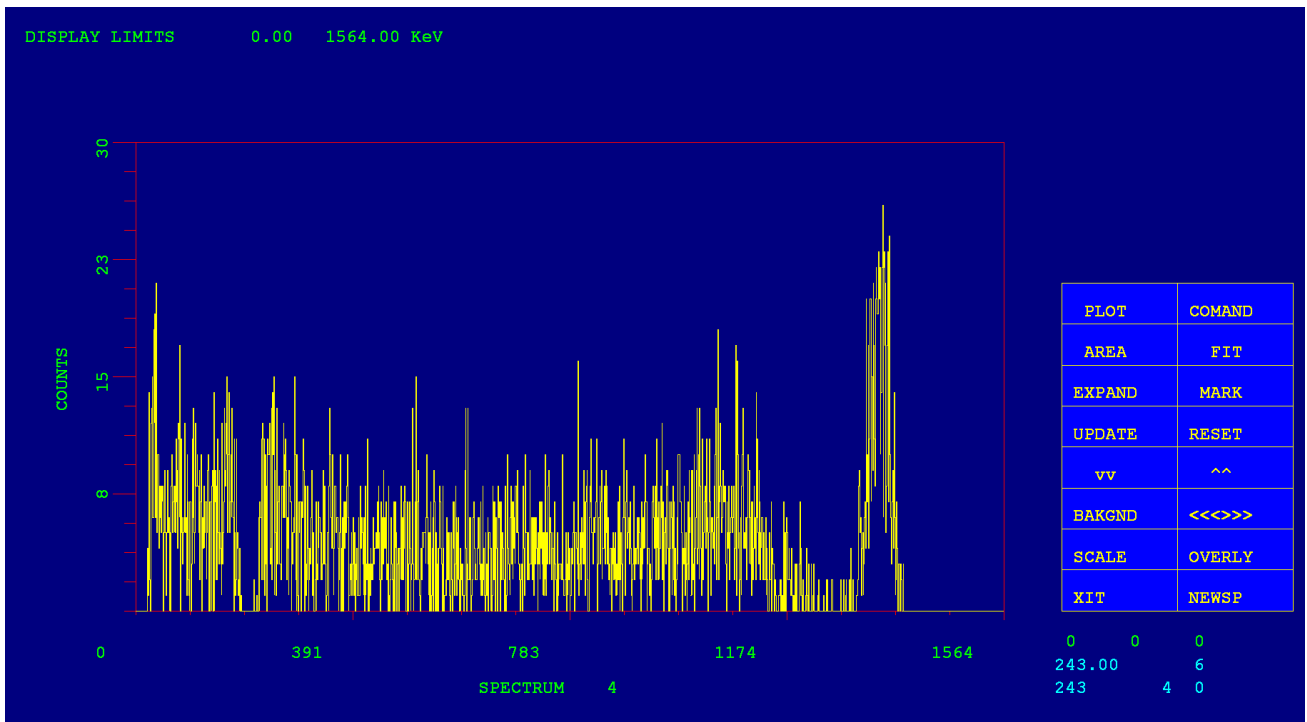
- (iii) 81 keV is in coincidence with 53 keV, 303 keV, 79 keV, 276 keV, 356 keV
- (iv) 353 keV is in coincidence with 53 keV, 81 keV.
- (v) 356 keV is in coincidence with 81 keV.
- (vi) 276 keV is in coincidence with 81 keV.
- (vii) 303 keV is in coincidence with 81 keV, 53 keV.

Reason In the source we have Ba^{133} which undergoes electron capture and forms Cs^{133} . If it is produced in $1/2^+$ state then it will emit 53 keV gamma which will be correlated with 303 keV and 81 keV in a decay chain which is quite close to our detection. In another decay chain we have 53 keV, 223 keV, 80 keV, 81 keV correlated gammas. As well as the $1/2^+$ state also can decay through 276 keV and then 81 keV in a decay chain which is exact to our observation. In another decay chain the $1/2^+$ state can also decay through 356 keV and 81 keV correlated gammas which is also exact with the observation. Also, if the Ba^{133} decays to $3/2^+$ state of Cs^{133} then it can decay through 223 keV and 161 keV correlated gammas.

Also, we have noticed that there are two more gammas of energy 1117 keV and 934 keV which are in coincidence with 211 keV gamma. Actually, the Compton edge of 1173 keV and 1332 keV gammas of Co^{60} occur at 958 keV and 1118 keV and the scattered photons are detected at the backscatter peak at energies 209 keV and 214 keV. Our detector detects these two gammas as one peak at 211 keV and since the backscatter peak and the Compton edge are timely correlated, we got 1117 keV and 934 keV in coincidence with the 211 keV peak.



(a) 1173 gated at 90°



(b) 1173 gated at 180°

Figure 4.5: Angular distribution plots of 1173 keV transition gated at 90° and 180° .

4.3.1 Angular distribution plots gated at 135° for various transitions.

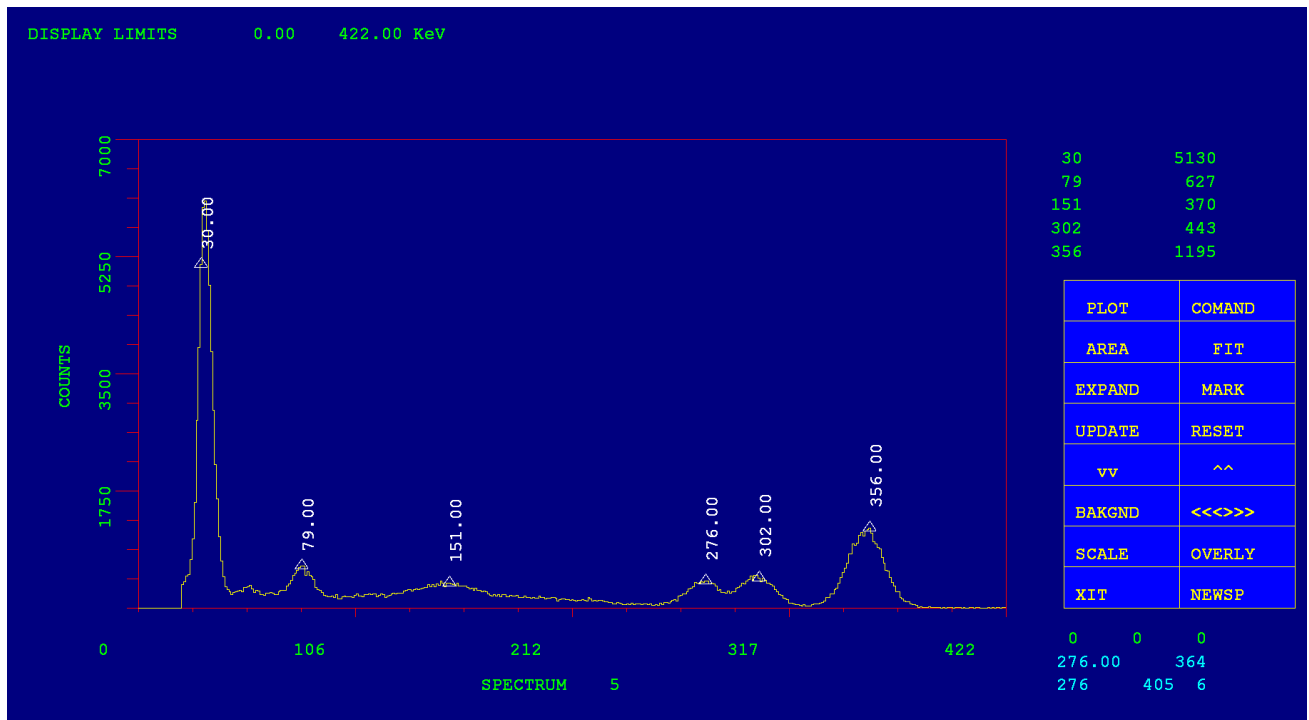


Figure 4.6: 81 keV gated at 135°

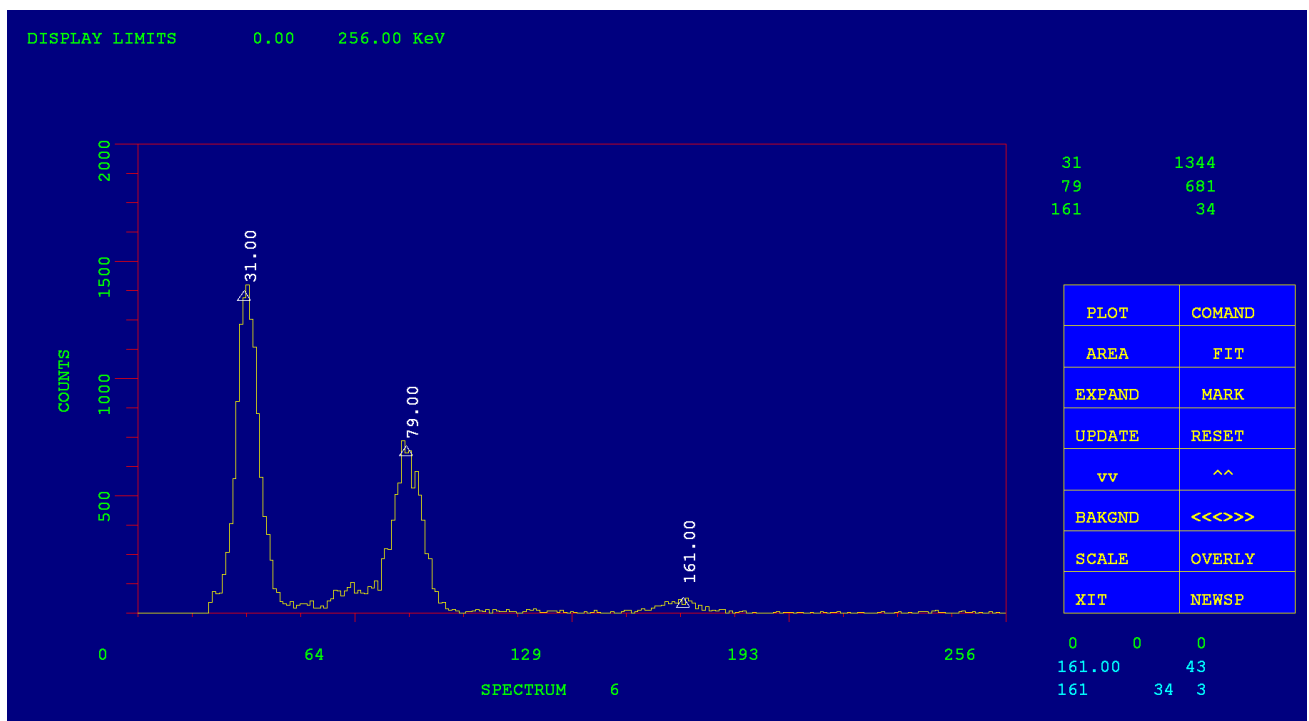


Figure 4.7: 276 keV gated at 135°

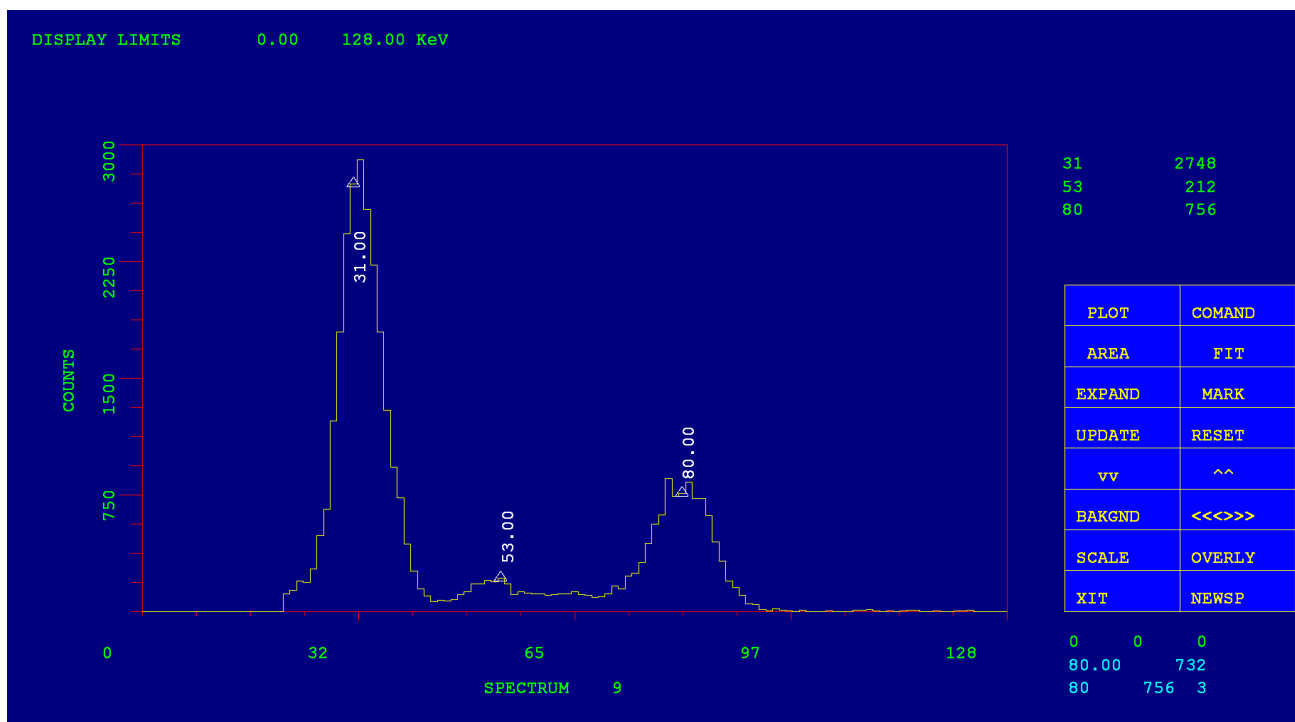


Figure 4.8: 303 keV gated at 135°

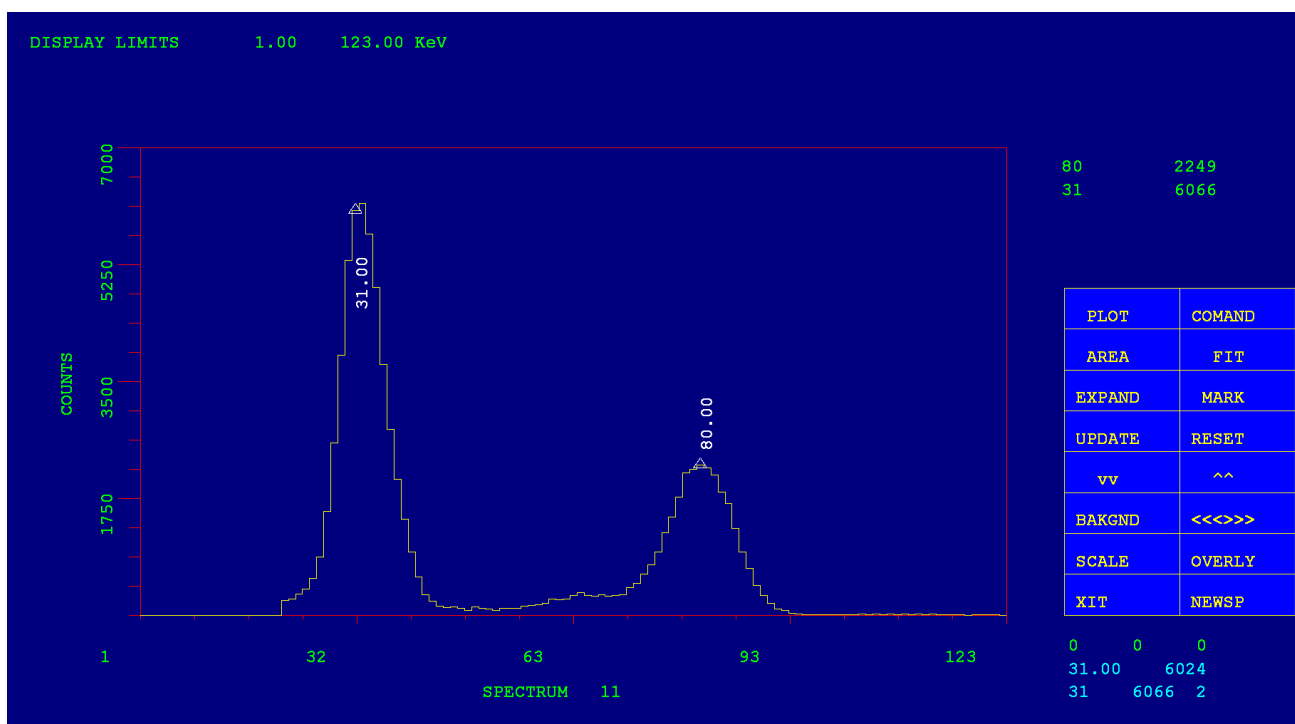


Figure 4.9: 356 keV gated at 135°

4.4 Discussion

There are actually 79 keV and 80 keV photopeaks in the level scheme which are not resolvable due to the poor energy resolution of our detector.

In the singles mode we have observed that there is another peak at 383 keV which is absent in coincidence mode. Actually, this is a photopeak of Ba^{133} which is a purely uncorrelated gamma. that's why we don't get it in the coincidence mode.

Experiment 5

Identification of elements and determination of their relative amount present in a given sample using X-Ray Fluorescence (XRF) technique

5.1 Objectives

1. To calibrate the XRF setup using known targets.
2. To identify the elements present in an unknown target and determine their relative amount.

5.2 Theory

X-rays can be produced by a radioactive source, an X-ray tube and by synchrotron radiation. In the present experiment, we use an X-ray tube. A typical X-ray tube is equipped with a heating filament as a cathode. Due to Joules heating by the electrical current through the filament electrons are emitted and become accelerated in a high electric field towards an anode. X-rays are produced in the target. There are two types of X-rays.

1. **Continuous X-rays:-** The broad-wavelength band of radiation is called continuum also known as Bremsstrahlung. It arises due to deceleration of the accelerated electrons within the target material due to their interaction with electrons and nucleus in the target element.
2. **Characteristic X-ray:-** Characteristic X- rays are produced when high energy electrons knock out the inner shell electrons from the atoms of the target material and to fill that vacancy electrons from higher shells jumps to the vacant states and emits a photon whose energy is equal to the energy gap between the two levels.

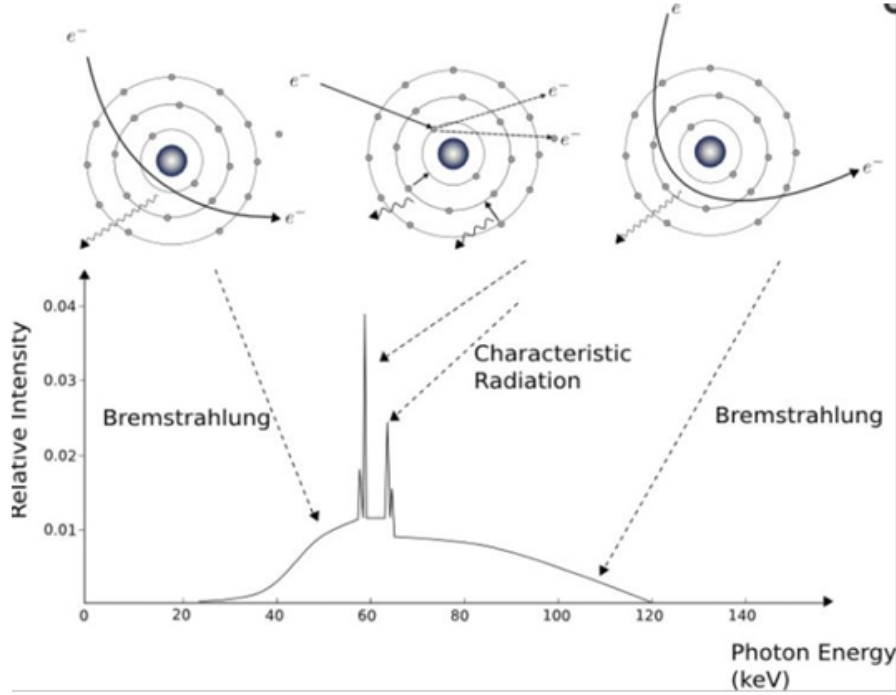


Figure 5.1: Production of characteristic and continuous X-rays

5.3 Principle of the X-ray fluorescence process:

If the primary energy of X-rays is equal to or is larger than the binding energy of an inner shell electron it is likely that electrons will be ejected and consequently vacancies are created. The hole state has certain life time and becomes refilled again. The transition of the excited atom into a state with lower energy occurs via two competitive processes, the above mentioned photoelectric and Auger effects. In the photoelectric effect, the recombination is accompanied by a transfer of electrons from the outer shells with energy E_m into the inner shells with energy E_n filling the vacancies. This process induces the emission of a characteristic X-ray (fluorescence) photon with energy $h\nu = E_m - E_n$. Therefore the energy of these secondary X-rays is the difference between the binding energies of the corresponding shells. The excited atom can also recombine by emission of Auger electrons, instead of characteristic X-rays, via the Auger effect. The probability that characteristic X-rays will be emitted - and not an Auger electron- varies from one element to another and is described as the fluorescence yield. For elements of low atomic numbers, the Auger effect dominates, whereas emission of characteristic X-rays is more likely for heavy elements. Each element has its unique characteristic energy spectrum (Fluorescence spectrum) composed by the allowed transitions of the specific atom in the result of X-ray excitation. XRF technique consists on the study of the produced characteristic spectrum.

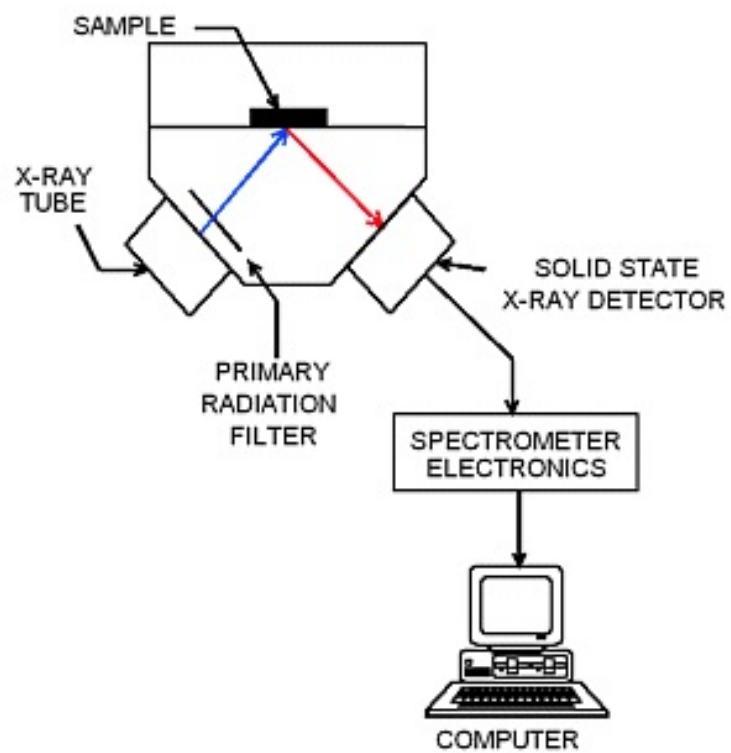


Figure 5.2: Block diagram of XRF experimental setup.

5.4 Experimental data

Data for channel calibration is shown below. We have used silver and copper targets.

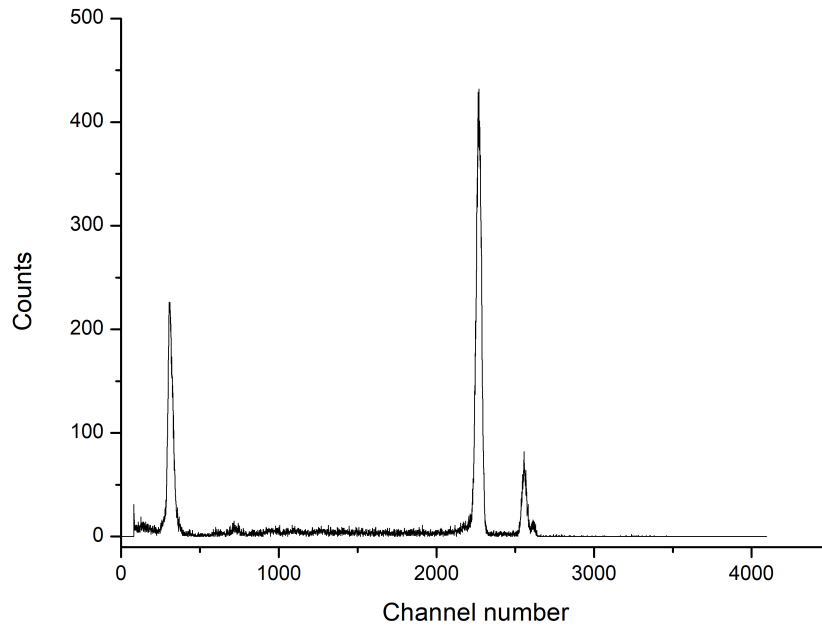


Figure 5.3: (a) Spectrum of Ag

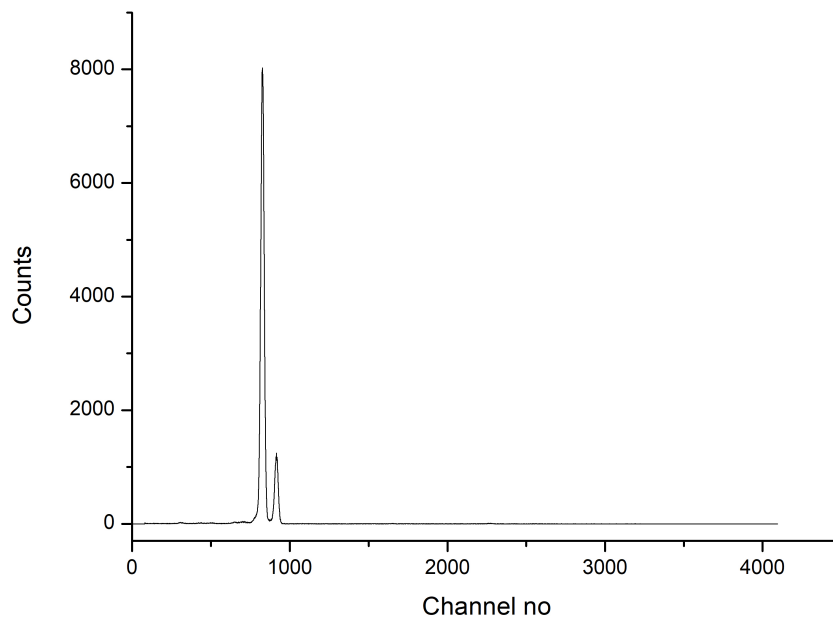


Figure 5.4: (b) Spectrum of Cu

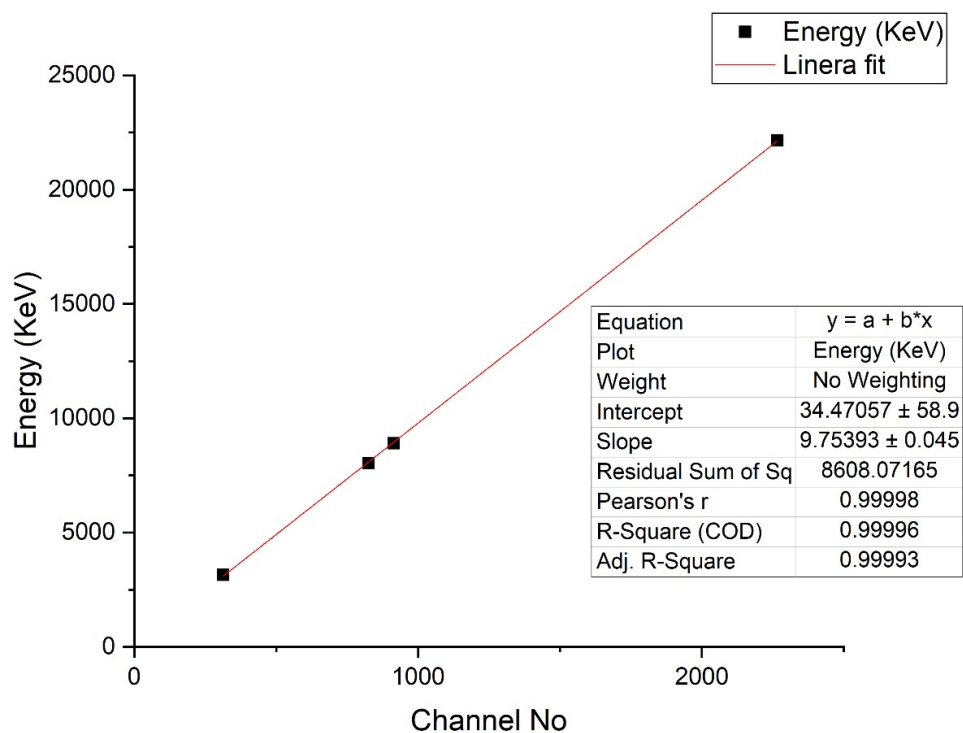


Figure 5.5: Calibration graph of XRF energy

We see that the slope of the graph is 9.75 ± 0.04 and the intercept is $(34.47 \pm 58.90)\text{keV}$. Now from this we can determine the energies corresponding to the X-ray peaks of the unknown samples.

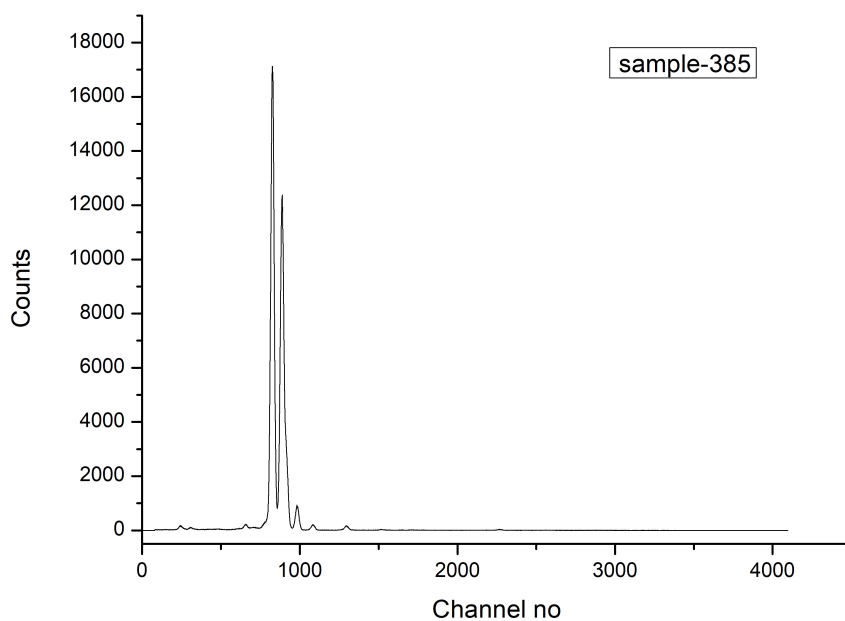


Figure 5.6: (a) X rays spectrum for Unknown sample 385

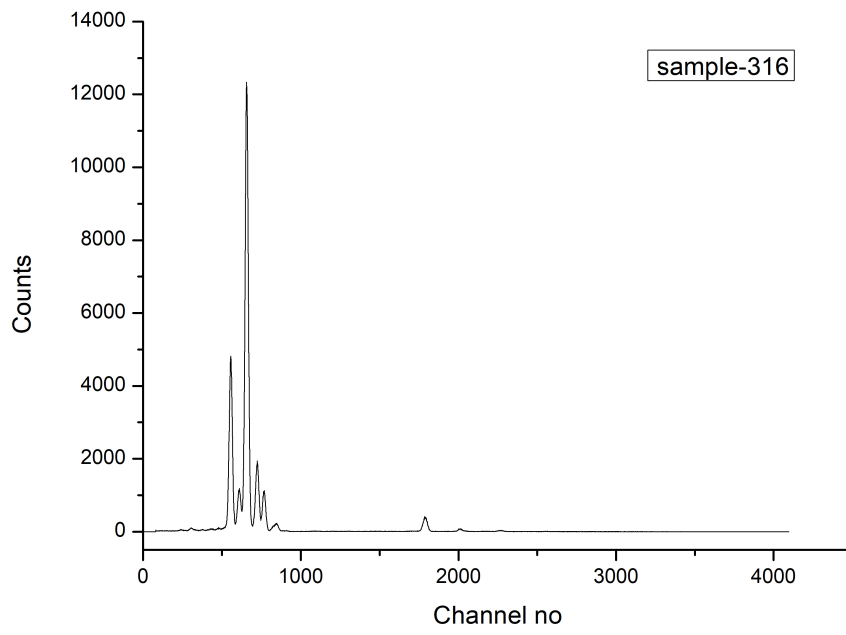


Figure 5.7: (b) X rays spectrum for Unknown sample 316

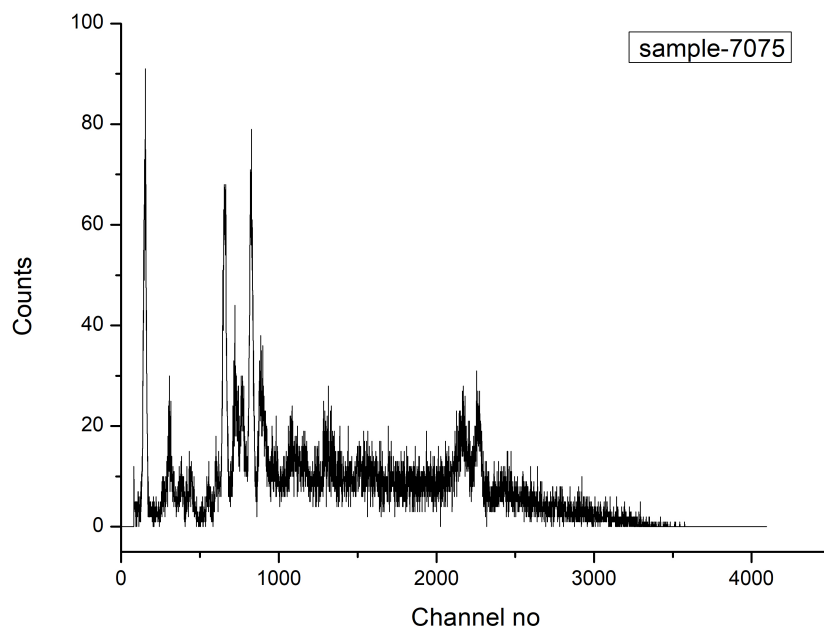


Figure 5.8: (c) X rays spectrum for Unknown sample 7075

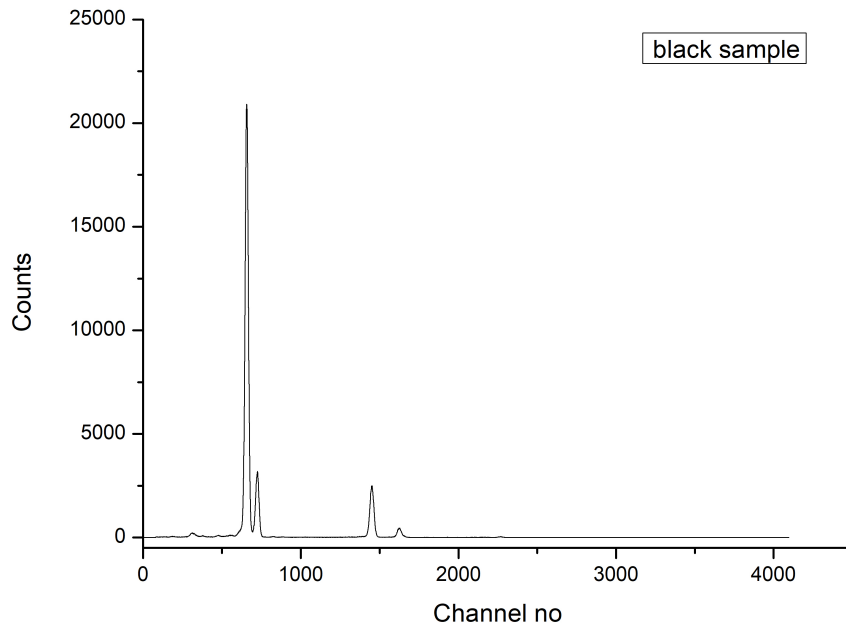


Figure 5.9: (d) X rays spectrum for Unknown sample Black

Now we have to consider the efficiency of the detector for different energies. The plot of actual efficiency vs energy is shown below.

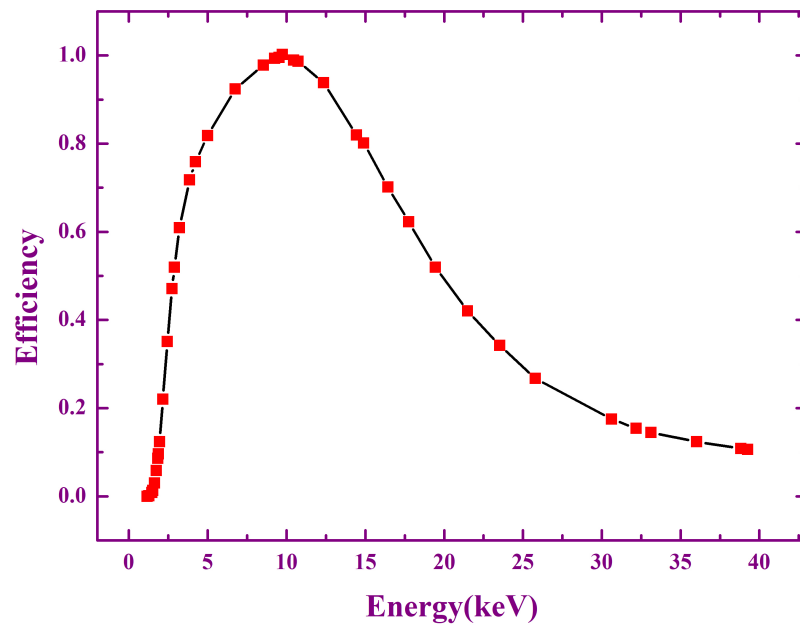


Figure 5.10: Efficiency vs Energy calibration

5.5 Result

Table 5.1: XRF Data for Sample 385

Channel	Energy (eV)	Exp. Count	Efficiency	Actual Count	Element	X-ray Peak	Peak Energy (eV)	Rel. Abundance (%)
244	2420.00	4128	0.35	11794	16 S	$K\beta_1$	2464.04	1.52
308	3043.00	2132	0.57	3740	18 Ar	$K\alpha_1$	2957.00	0.48
825	8079.09	416807	0.96	434174	29 Cu	$K\alpha_1$	8047.78	56.16
887	8682.97	285300	0.98	291122	30 Zn	$K\alpha_1$	8638.86	40.47
982	9608.27	21769	1.00	21769	30 Zn	$K\beta_1$	9572.00	
1081	10572.53	5582	0.99	5638	33 As	$K\alpha_1$	10543.72	0.72
1295	12656.89	4432	0.93	4766	36 Kr	$K\alpha_1$	12649.00	0.61

Table 5.2: XRF Data for Sample 316

Channel	Energy (eV)	Exp. Count	Efficiency	Actual Count	Element	X-ray Peak	Peak Energy (eV)	Rel. Abundance (%)
554	5439.55	107957	0.85	127008	24 Cr	$K\alpha_1$	5414.72	20.46
608	6965.51	21922	0.87	25197	27 Co	$K\alpha_1$	6930.32	4.06
655	6423.29	296980	0.90	329977	26 Fe	$K\alpha_1$	6403.84	61.04
722	7075.87	45472	0.93	48894	26 Fe	$K\beta_1$	7057.98	
766	7504.43	27096	0.45	60213	28 Ni	$K\alpha_1$	7478.15	10.80
839	8215.45	6592	0.96	6866	28 Ni	$K\beta_1$	8264.66	
1787	17448.97	12635	0.64	19742	42 Mo	$K\alpha_1$	17479.39	3.18
2265	22104.60	1067	0.40	2667	47 Ag	$K\alpha_1$	22162.92	0.42

Table 5.3: XRF Data for Sample 7075

Channel	Energy (eV)	Exp. Count	Efficiency	Actual Count	Element	X-ray Peak	Peak Energy (eV)	Rel. Abundance (%)
149	1494.85	1442	0.02	72100	13 Al	$K\alpha_1$	1486.7	93.45
655	6423.29	1417	0.9	1574	26 Fe	$K\alpha_1$	6403.86	2.04
824	8069.35	1346	0.96	1394	29 Cu	$K\alpha_1$	8047.78	1.80
2166	21140.43	353	0.43	811	46 Pd	$K\alpha_1$	21177.1	1.05
2263	22085.21	495	0.39	1269	47 Ag	$K\alpha_1$	22162.92	1.64

Table 5.4: XRF Data for Sample Black

Channel	Energy (eV)	Count	Efficiency	Actual Count	Element	X-ray Peak	Peak Energy (eV)	Percent Yield (%)
656	6433.00	517460	0.90	574955	26 Fe	$K\alpha_1$	6403.84	97.33
722	7075.80	75713	0.93	81411	26 Fe	$K\beta_1$	7057.00	
307	3033.70	4652	0.57	8161	18 Ar	$K\alpha_1$	2957.70	1.21
329	3248.05	2386	0.61	3911	19 K	$K\alpha_2$	3311.10	0.58
376	3705.83	1331	0.71	1874	20 Ca	$K\alpha_1$	3691.68	0.27
824	8069.35	719	0.96	748	29 Cu	$K\alpha_1$	8047.78	0.11
885	8663.49	412	0.97	424	30 Zn	$K\alpha_1$	8638.86	0.06
475	4670.09	1327	0.79	1679	22 Ti	$K\alpha_1$	4510.84	0.24
551	5410.33	959	0.84	1141	24 Cr	$K\alpha_1$	5414.72	0.16

5.6 Discussion

The X-ray Fluorescence (XRF) analysis of four different samples—Sample 365, Sample 316, Sample 7075, and Sample bg 901—has provided comprehensive insights into their elemental composition and potential alloy classifications:

Sample 365: This sample is dominated by Copper (56.16%) and Zinc (40.47%), indicating that it is most likely a brass alloy. Minor elements such as Sulfur, Arsenic, and noble gases are present in trace amounts, possibly due to alloying or environmental exposure.

Sample 316: Characterized by high concentrations of Iron (61.04%), Chromium (20.46%), and Nickel (10.80%), this sample matches the typical composition of AISI 316 stainless steel, a corrosion-resistant alloy commonly used in marine and industrial environments. The presence of Molybdenum (3.18%) further supports this identification.

Sample 7075: Predominantly composed of Aluminum (93.45%), this sample aligns with the composition of 7075 aluminum alloy, known for its high strength-to-weight ratio. The presence of small amounts of Iron, Copper, Palladium, and Silver suggests tailored alloying for enhanced mechanical properties or specific industrial uses.

Sample bg 901: With Iron accounting for 97.33% of the composition, this sample is essentially pure iron or a low-alloy steel. Minor traces of Argon, Potassium, Calcium, Titanium, Chromium, Copper, and Zinc are present, likely as impurities or environmental residues.

Overall Interpretation: The XRF analysis successfully differentiated between various metallic materials:

Brass (Sample 365)

Stainless Steel (Sample 316)

Aluminum Alloy (Sample 7075)

High-Purity Iron (Sample bg 901)

The results demonstrate the effectiveness of XRF in qualitative and semi-quantitative elemental analysis, enabling accurate material identification and quality control in metallurgical studies.